EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 1 – Lecture 1

Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **C-509**

Ohm's Law

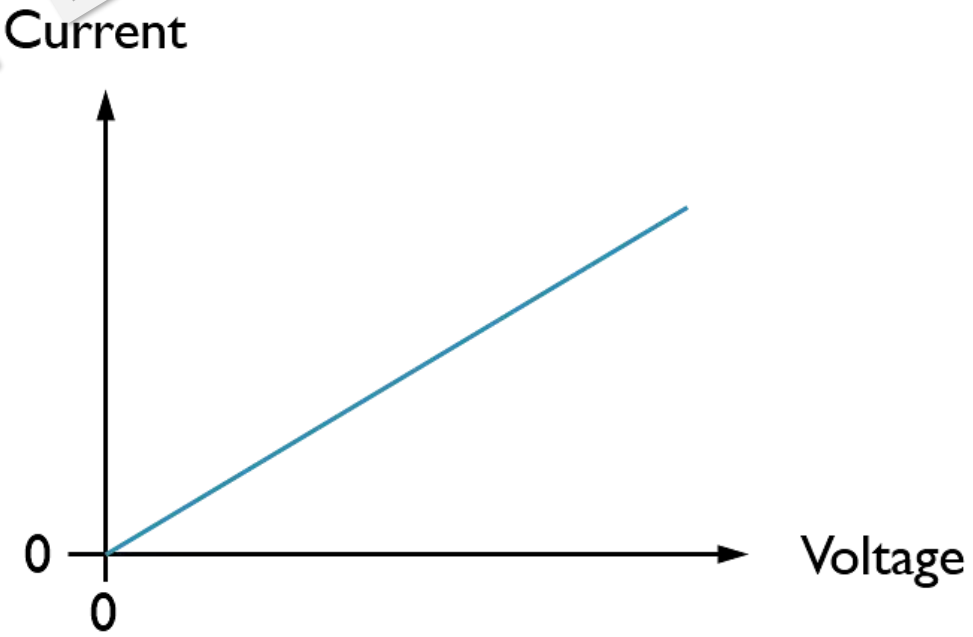
Voltage (V) between two terminals of a current-carrying conductor is directly proportional to the current, I flowing through it, provided the temperature and other physical conditions do not change.

$$V \propto I$$

$$V = I \times R$$

The resistance (R) is

- directly proportional to length of material (l)
- inversely proportional to area of cross section of material (A)
- $R = \rho\, l / A$; ρ is a constant called the resistivity of the material (SI unit: Ω m)



Resistance of pure metals increases with increase in temperature.

Dr. Arindam Dutta

Ohm's Law

Q: A silver wire is 5 cm long and has an inner diameter of 0.5 cm. What is the current flowing through the conductor if the potential difference between the two ends of the conductor is 10 V? Specific resistance (or resistivity) of silver is 0.016 μΩ m.

A: $l = 5 \text{ cm} = 0.05 \text{ m}$

Diameter (d) = 0.5 cm = 0.005 m

$A = \pi \times (d/2)^2 = 1.96 \times 10^{-5} \text{ m}^2$

$\rho = 1.6 \times 10^{-8} \ \Omega\text{m}$

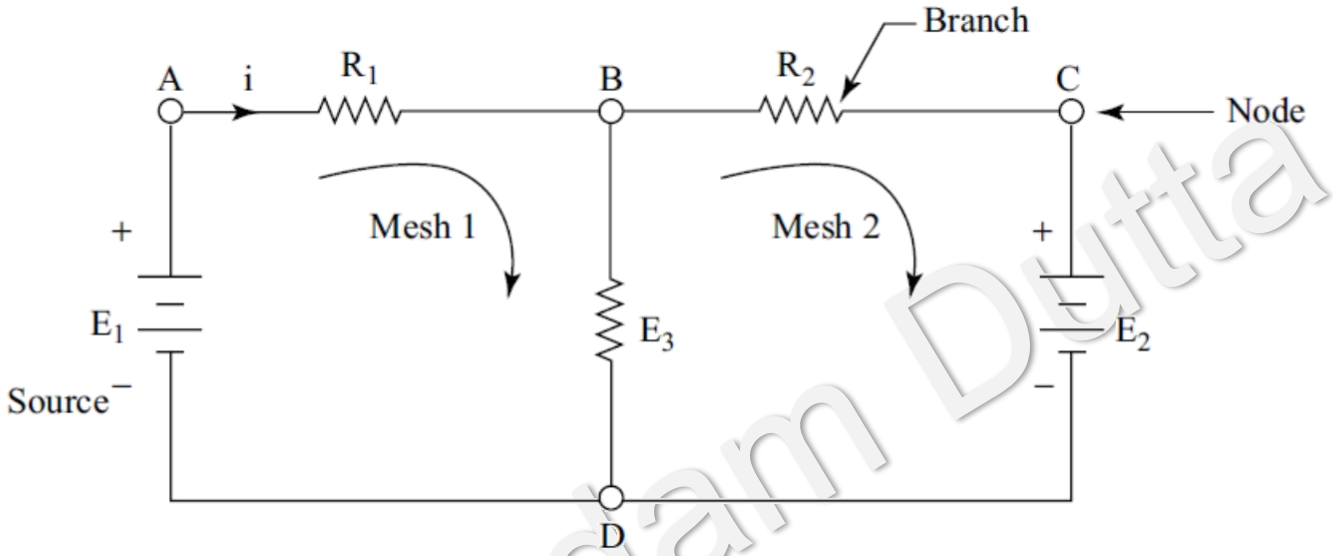
$R = \rho \, l / A = 4.08 \times 10^{-5} \ \Omega$

V = 10 V

Therefore, $I = V / R = 245098 \text{ A} = 245 \text{ kA}$

Dr. Arindam Dutta

DC Networks

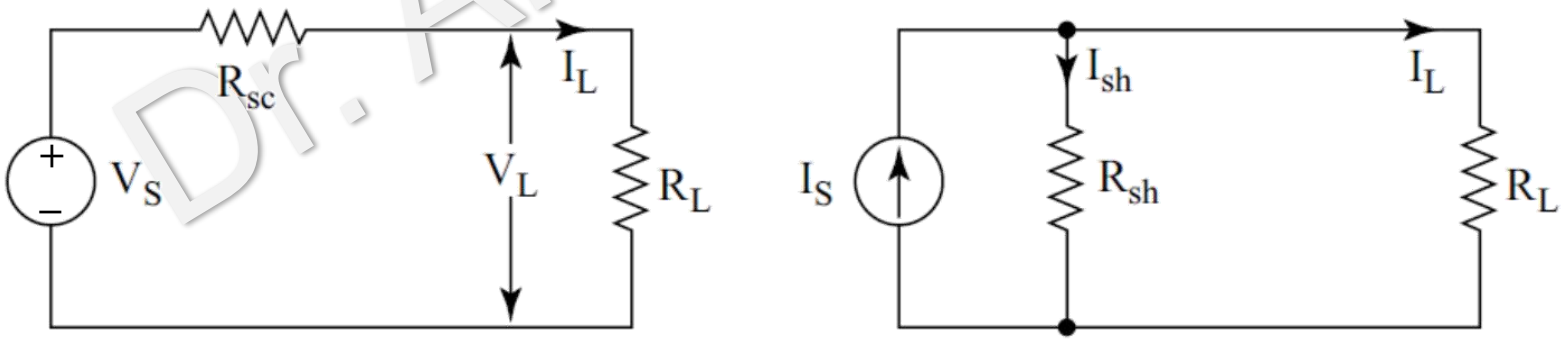


- No. of nodes, N = 4 (A, B, C, D)
- No. of branches, B = 5 (AB, BC, BD, CD, AD)
- Independent meshes, M = B – N + 1 = 5 – 4 + 1 = 2 (ABDA, BCDB)
- No. of loops = 3 (ABDA, BCDB and ABCDA). It is seen that a loop ABCDA encloses two meshes, i.e., mesh 1 and mesh 2.

Dr. Arindam Dutta

Voltage and Current Sources

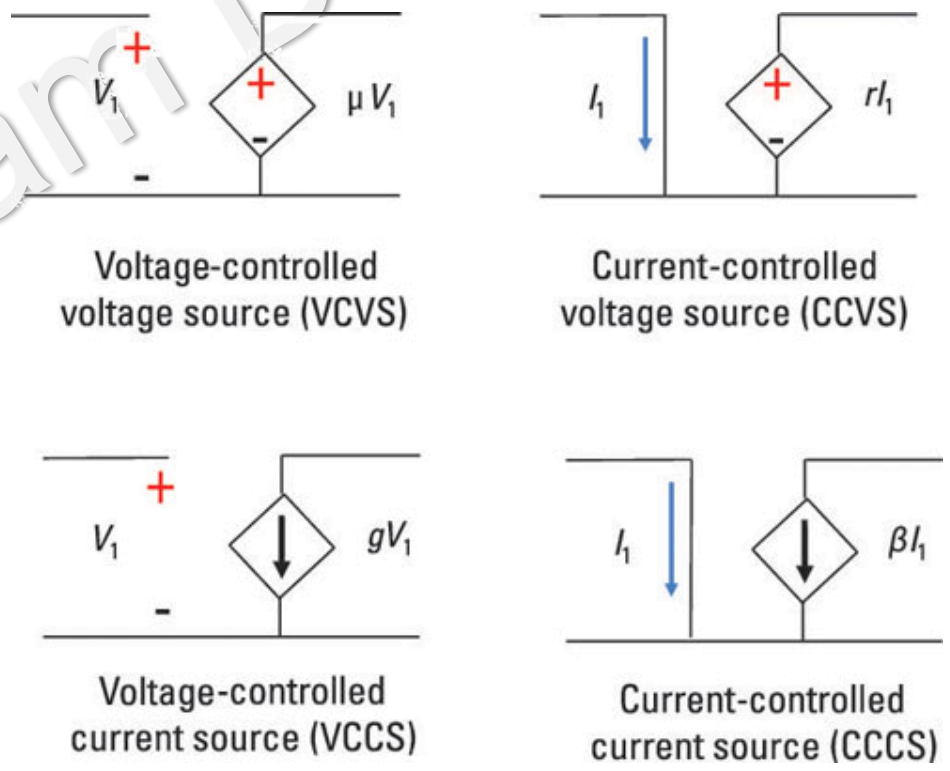
- A practical voltage source has small internal resistance in series while a current source has some high internal resistance in parallel
- For ideal voltage source $R_{se} = 0$
- For ideal current source $R_{sh} = \infty$
- Practical voltage source is shown as an ideal voltage source in series with a resistance (R_{se})
- Practical current source is shown as an ideal current source in parallel with its internal resistance (R_{sh})



Dr. Arindam Dutta

Independent and Dependent Sources

- The magnitude of an independent source (represented by a circle) does not depend upon the current in the circuit or voltage across any other element in the circuit
- The magnitude of a dependent source (represented by a diamond) gets changed due to some other current or voltage in the circuit
- There are four kinds of dependent sources:
  - voltage-controlled voltage source (vcvs)
  - current-controlled current source (cccs)
  - voltage-controlled current source (vccs)
  - current-controlled voltage source (ccvs)

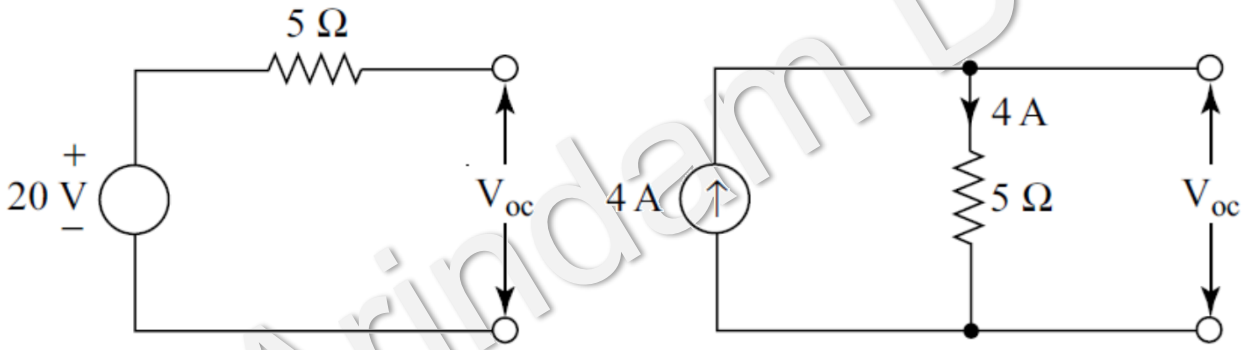


Dr. Arindam Dutta

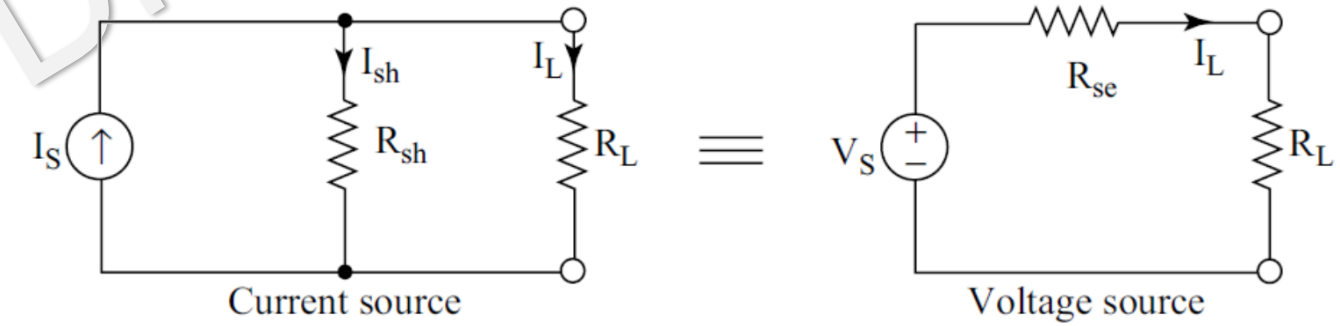
Source Transformation

- A voltage source is equivalent to a current source and vice-versa if
  - they produce equal values of I_L and V_L when connected to the load R_L
  - they provide the same open-circuit voltage and short-circuit current

- If voltage source is converted into current source, we consider the short circuit current equivalence then $I_s = V_s / R_{se}$



- If current source converted into voltage source, we consider the open-circuit voltage equivalence, then, $V_s = I_s R_{sh}$



Dr. Arindam Dutta

Source Transformation

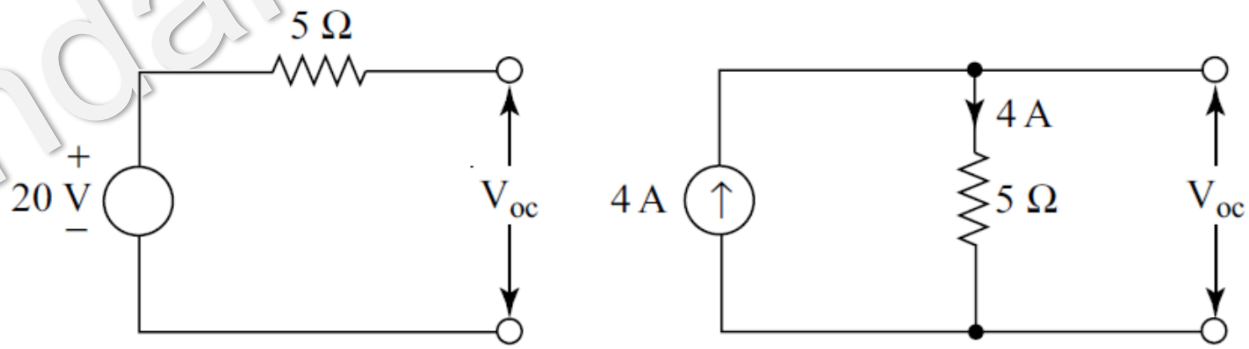
Q: Convert a voltage source of 20 V with internal resistance of 5 Ω into an equivalent current source.

A: $I_s = V_s / R_{se} = 4$ A

The internal resistance will be the same as R_{se}

The condition for equivalence is checked from the following conditions viz V_{oc} should be same and I_{sc} should also be same.

In first figure, $V_{oc} = 20$ V
In second figure, $V_{oc} = 4$ A × 5 Ω = 20 V



I_{sc} in first figure = 4 A
I_{sc} in second figure = 4 A

Thus, the two circuits are equivalent because the open circuit voltage and short circuit current are the same in both the circuits.

Dr. Arindam Dutta

Source Transformation

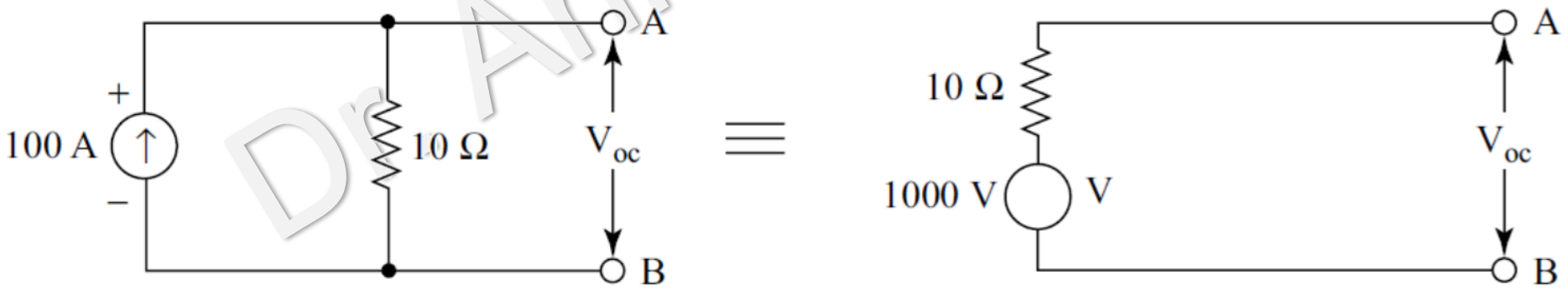
Q: Convert a current source of 100 A with internal resistance of 10 Ω into an equivalent voltage source.

A: For an equivalent voltage source,

$V = I \times R_{sh} = 100 \times 10 = 1000$ V

$R_{sh} = R_{se} = 10$ Ω in series

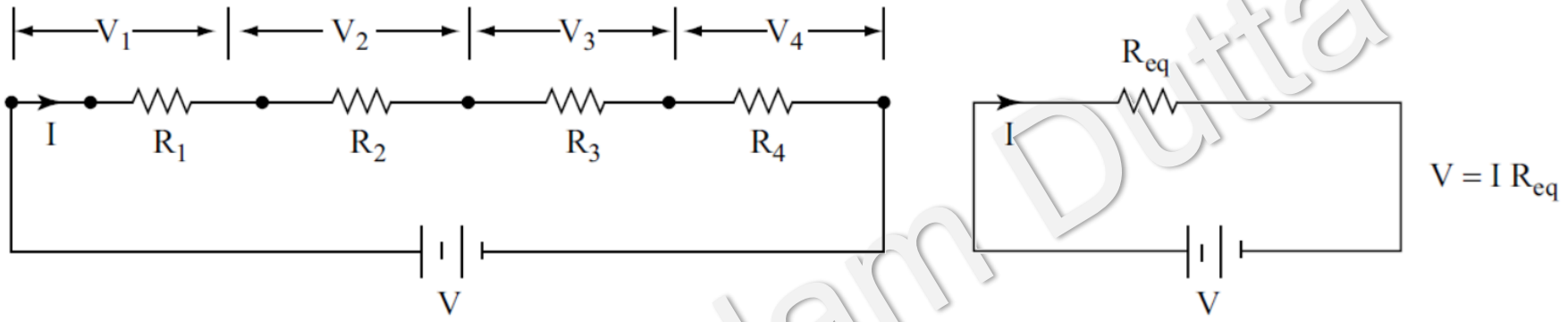
The open circuit voltage and short circuit current are the same in the two equivalent circuits.



Dr Arindam Dutta

Series Circuit

- Resistances are connected end to end across a source of supply



- Voltage V is connected across the resistors
- The same current is flowing through all the resistances,

$$V_1 = IR_1,\ V_2 = IR_2,\ V_3 = IR_3, \text{ and } V_4 = IR_4$$

$$V = V_1 + V_2 + V_3 + V_4$$

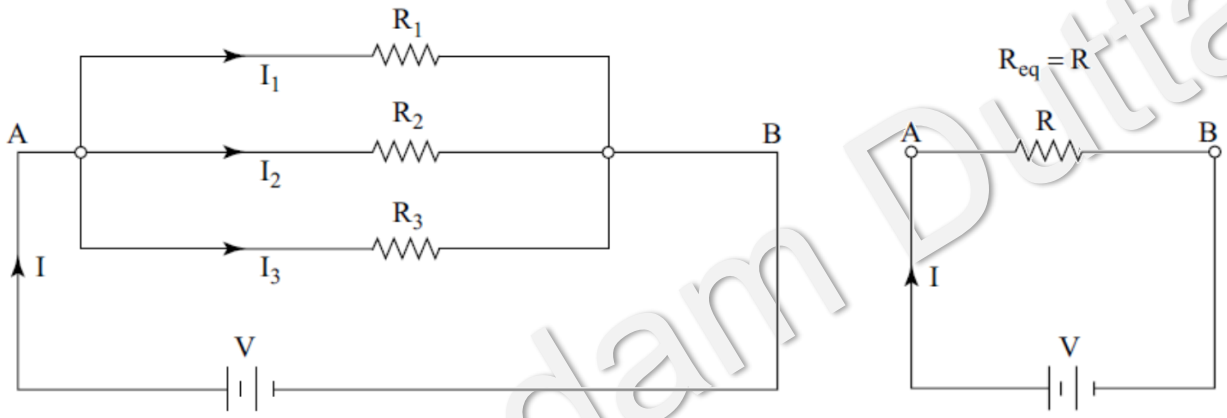
$$\text{or, } IR_{eq} = IR_1 + IR_2 + IR_3 + IR_4$$

$$\text{or, } R_{eq} = R_1 + R_2 + R_3 + R_4$$

Dr. Arindam Dutta

Parallel Circuit

- Resistors are connected in such a way that both the ends of individual resistors are connected together and two terminals are brought out for connection to other parts of a circuit



- Voltage V is connected across the three resistors and total current drawn is I

$$I = I_1 + I_2 + I_3 = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

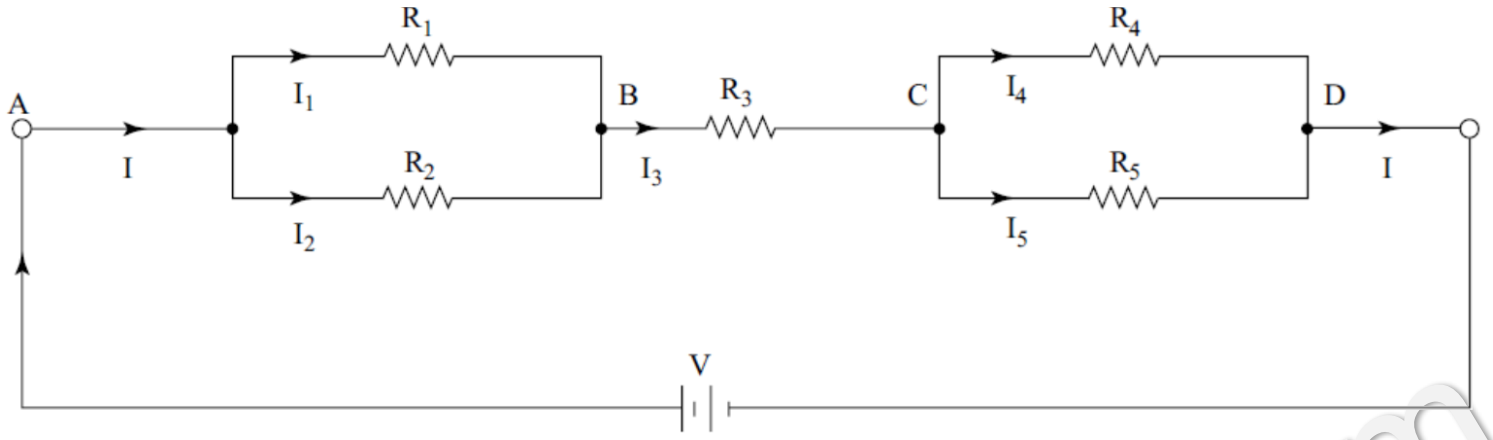
$$I = \frac{V}{R}$$

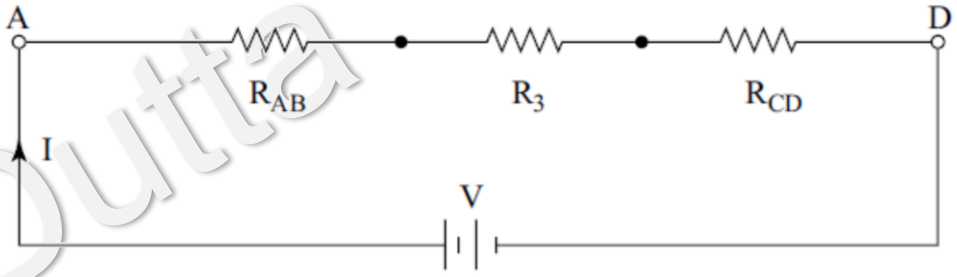
$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Dr. Arindam Dutta

Series-Parallel Circuit





$$\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{AB} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{1}{R_{CD}} = \frac{1}{R_4} + \frac{1}{R_5}$$

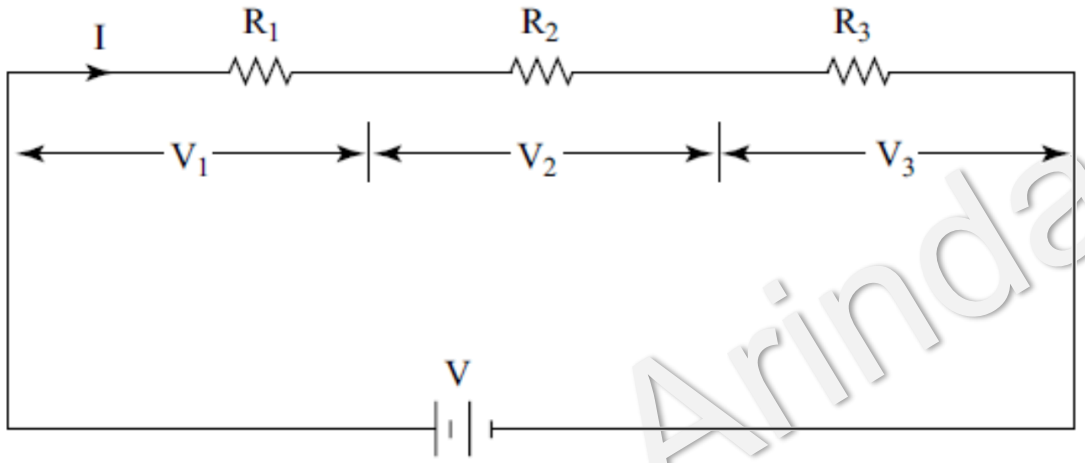
$$R_{CD} = \frac{R_4 R_5}{R_4 + R_5}$$

$$R = \frac{R_1 R_2}{R_1 + R_2} + R_3 + \frac{R_4 R_5}{R_4 + R_5}$$

Dr. Arindam Dutta

Voltage Divider Rule

For easy calculation of voltage drop across resistors in a series circuit, a voltage divider rule is used



$$I = \frac{V}{R_1 + R_2 + R_3}$$

$$V_1 = I\,R_1 = \frac{V}{R_1 + R_2 + R_3} \times R_1 = \frac{V}{R_T} \times R_1$$

$$R_T = R(\text{Total}) = R_1 + R_2 + R_3$$

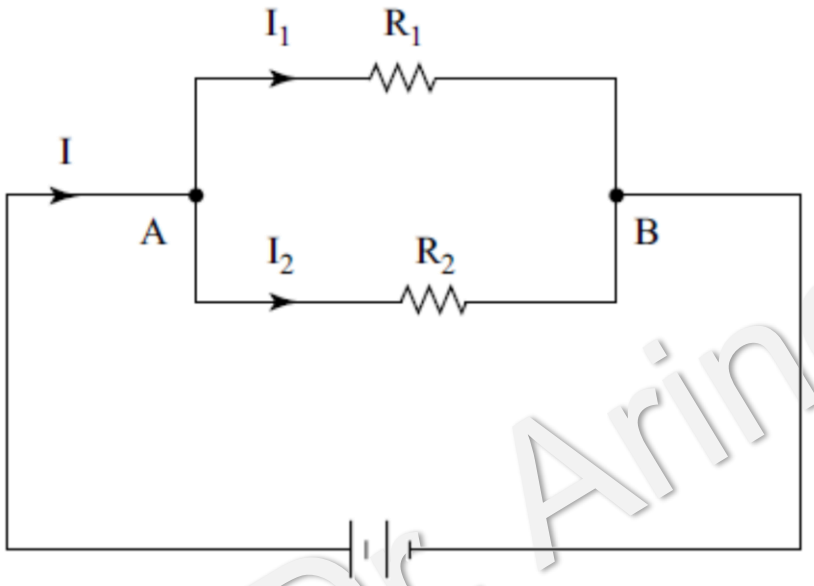
$$V_2 = I\,R_2 = \frac{V}{R_T} \times R_2$$

$$V_3 = I\,R_3 = \frac{V}{R_T} \times R_3$$

Dr. Arindam Dutta

Current Divider Rule

Similarly, current divider rule is used in parallel circuits to find the branch currents if the total current is known.



$$V_{AB} = I_1 R_1 = I_2 R_2$$

$$I = I_1 + I_2$$

$$I_1 R_1 = (I - I_1) R_2$$

$$I_1 (R_1 + R_2) = I R_2$$

$$I_1 = I \frac{R_2}{R_1 + R_2}$$

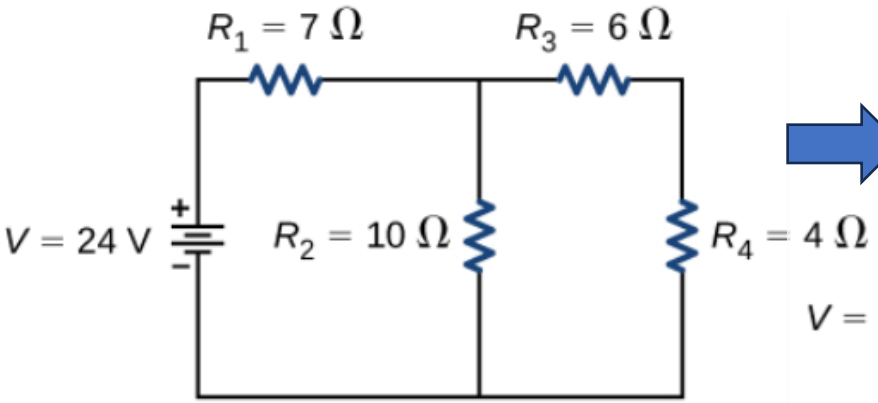
$$I_2 = I - I_1 = I - I \frac{R_2}{R_1 + R_2}$$

$$= I \left[1 - \frac{R_2}{R_1 + R_2} \right]$$

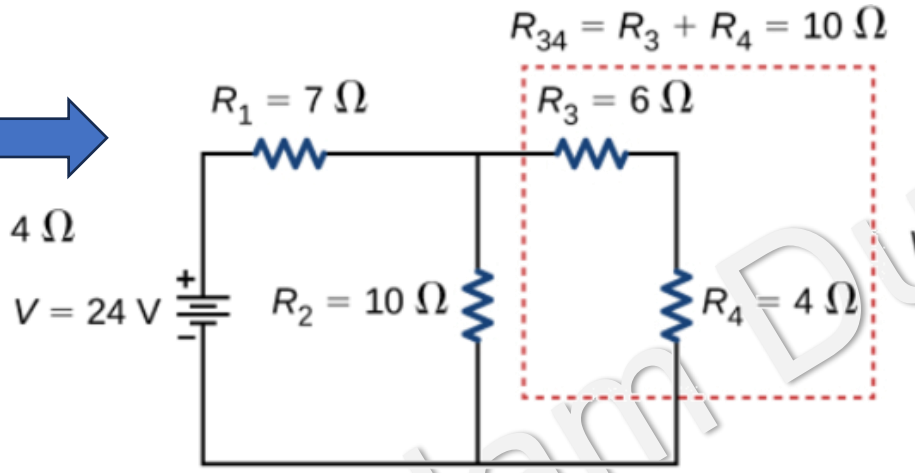
$$I_2 = I \frac{R_1}{R_1 + R_2}$$

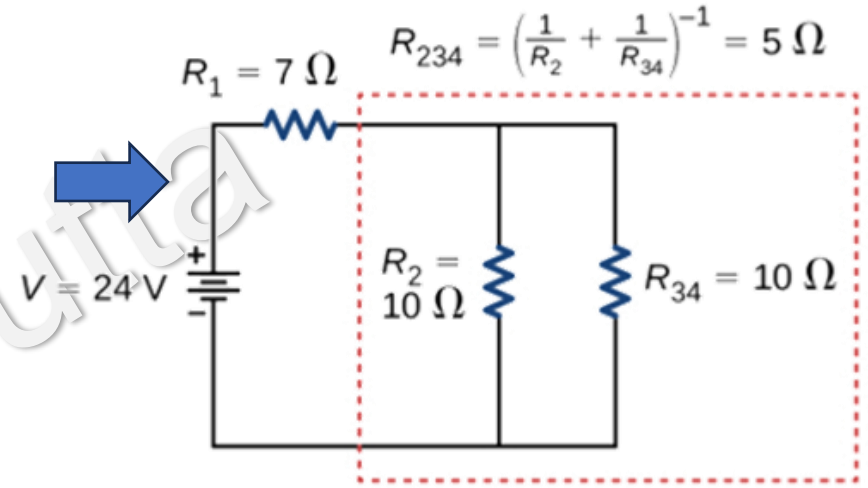
Dr. Arindam Dutta

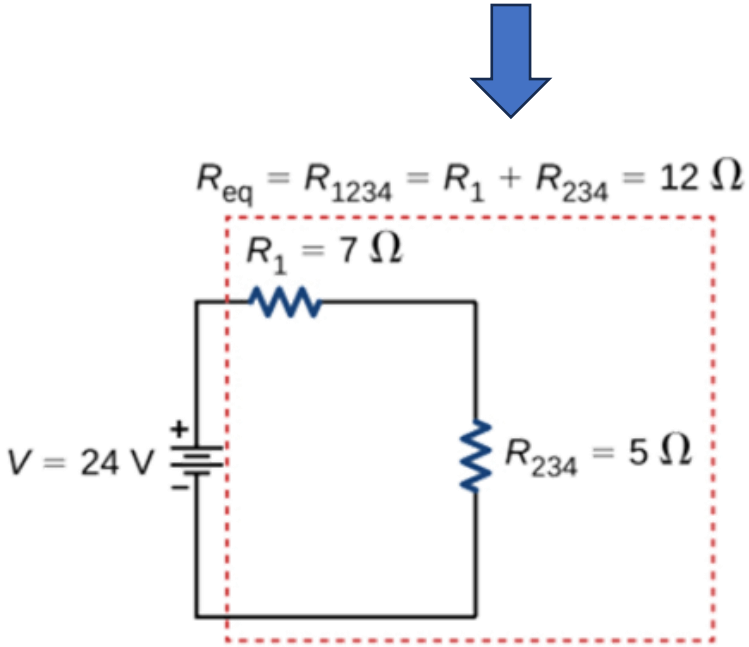
Problem



I = ?









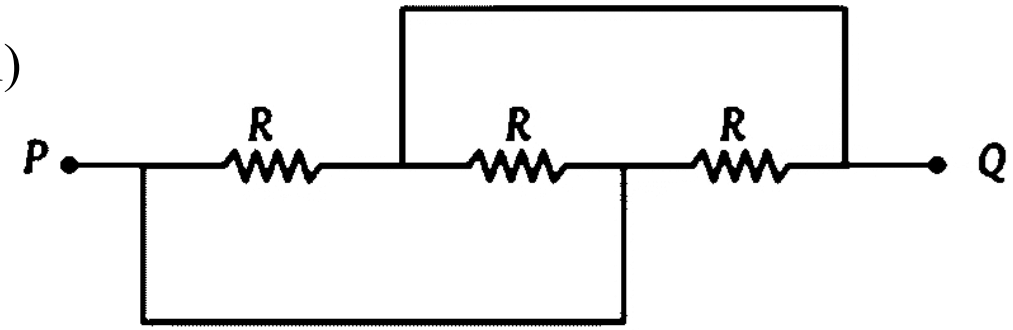
$I = V / R_{eq} = 2$ A

Dr. Arindam Dutta

Problems

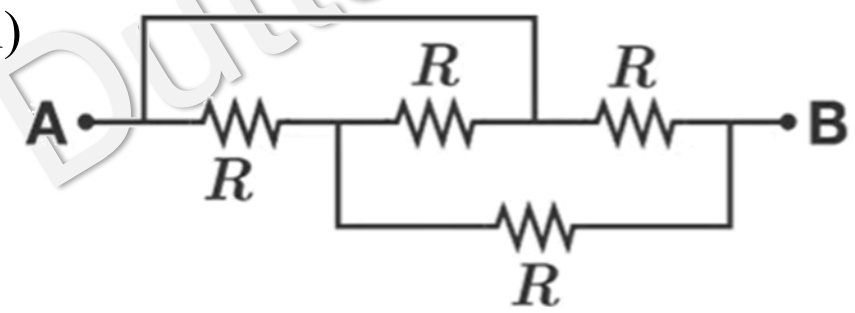
Calculate the equivalent resistances of the following circuits

(i)



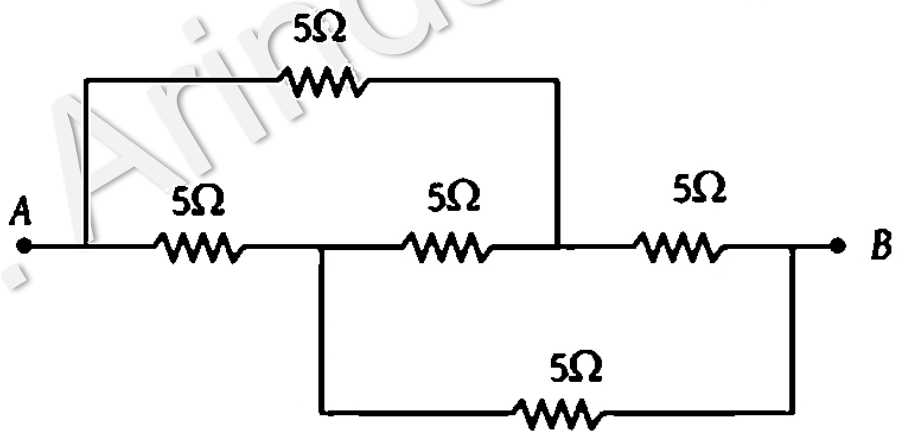
$R_{eq} = R/3$

(ii)



$R_{eq} = 3R/5$

(iii)

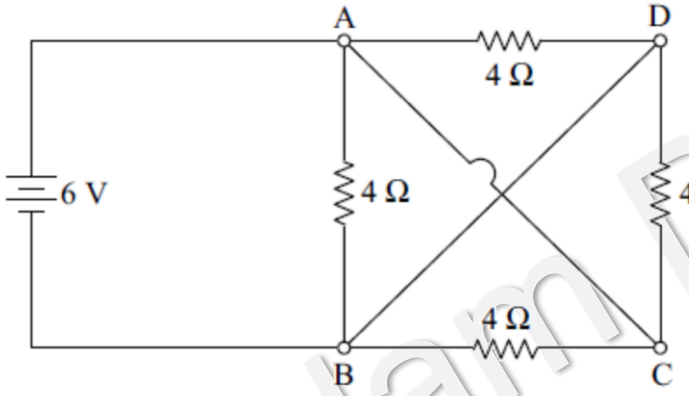


$R_{eq} = 5\ \Omega$

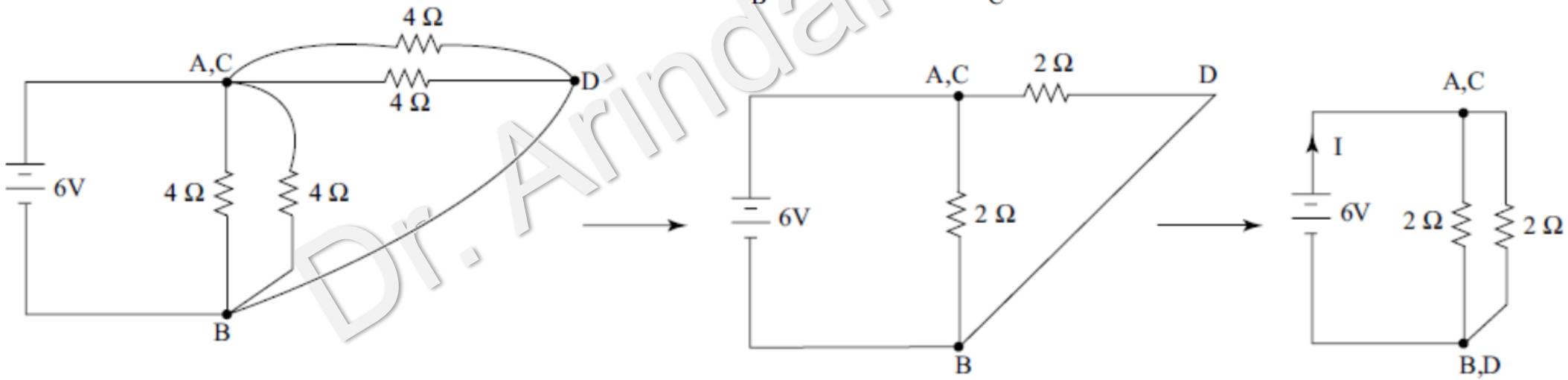
Dr. Arindam Dutta

Problem

Q: Calculate the current supplied by the battery in the network.



A:



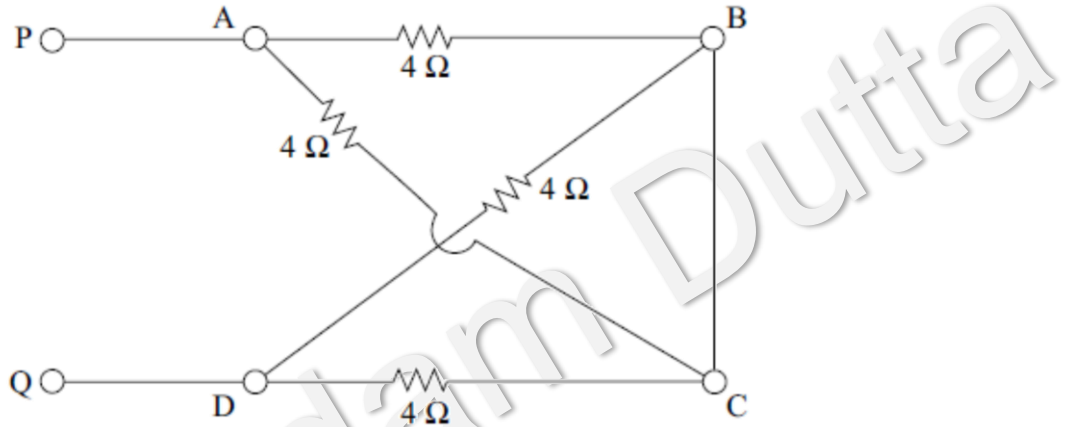
Paralleling the two 2 Ω resistors, $R_{eq} = 2 \times 2 / (2+2) = 1\ \Omega$. Therefore, the current I is calculated as (6 / 1) A = 6 A.

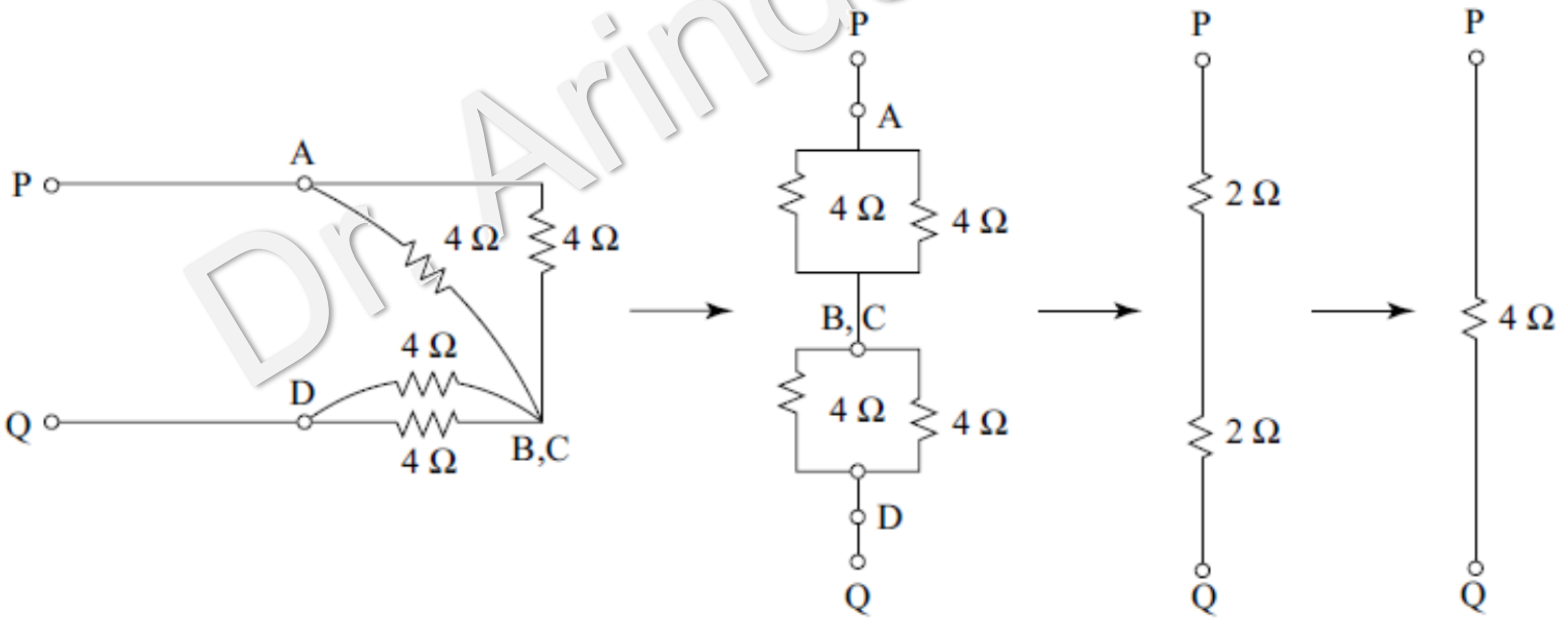
Dr. Arindam Dutta

Problem

Q: Calculate the resistance between the terminals P and Q of the network.

A:





Dr Arindam Dutta

EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 1 – Lecture 2

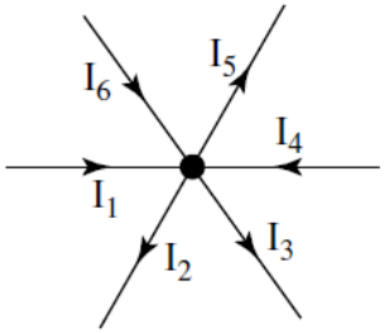
Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **C-509**

Kirchhoff's Current Law

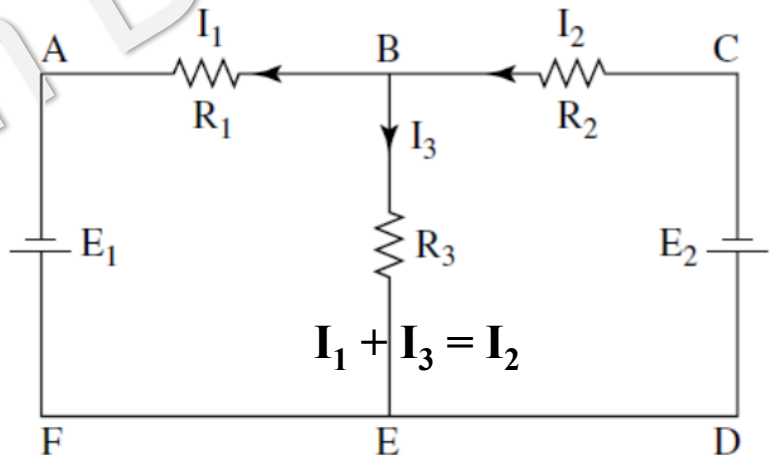
- This law is applied at any node of an electric network
- It states that the algebraic sum of currents meeting at a junction or a node in a circuit is zero
- Mathematically: $\sum_{j=1}^{n} \mathbf{I_j} = \mathbf{0}$, where n is the number of branches meeting at a node and I_j represents the current in the jth branch



or

$$I_1 + I_6 + I_4 - I_5 - I_3 - I_2 = 0$$

$$I_1 + I_6 + I_4 = I_5 + I_3 + I_2$$

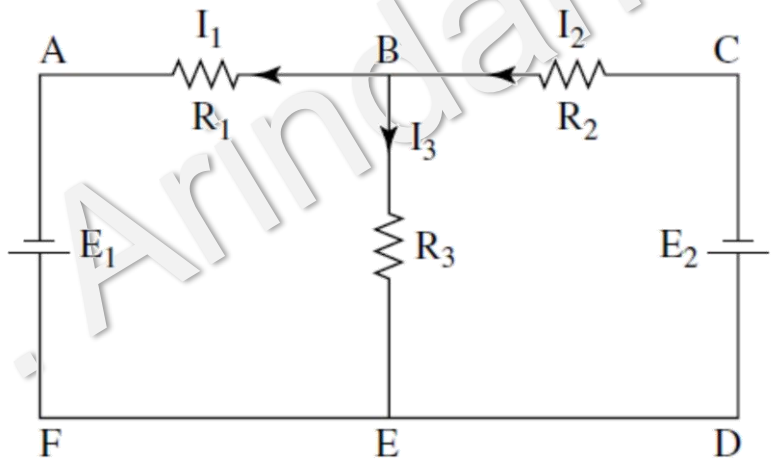


- The sum of current flowing towards a junction or a node is equal to the sum of currents flowing out of the junction
- The current entering the junction has been taken as positive while the currents leaving the junction have been taken as negative

Dr. Arindam Dutta

Kirchhoff's Voltage Law

- At any instant of time the algebraic sum of voltages in a closed loop is zero.
- In applying KVL in a loop or a mesh, a positive sign may be assigned to the rise in voltage and negative sign may be assigned to the fall or drop in voltage
- Mathematically: $\sum_{j=1}^{n} V_j = 0$, where V_j represents the voltages of all the branches in a mesh or a loop, i.e., in the jth element around the closed loop having n elements



- Applying KVL in mesh ABEFA and mesh CBEDC

Dr. Arindam Dutta

Kirchhoff's Voltage Law

- The voltage is taken as negative in the direction of the flow of current
- Voltage is taken as positive from the negative terminal of the source of EMF to the positive terminal
- ABEFA: starting from A, sum of voltage drops and voltage rise are equated to zero as

$$+ I_1 R_1 - I_3 R_3 - E_1 = 0$$

$$\text{or, } I_1 R_1 - (I_2 - I_1) R_3 - E_1 = 0$$

- Similarly for CBEDC:

$$- I_2 R_2 - I_3 R_3 - E_2 = 0$$

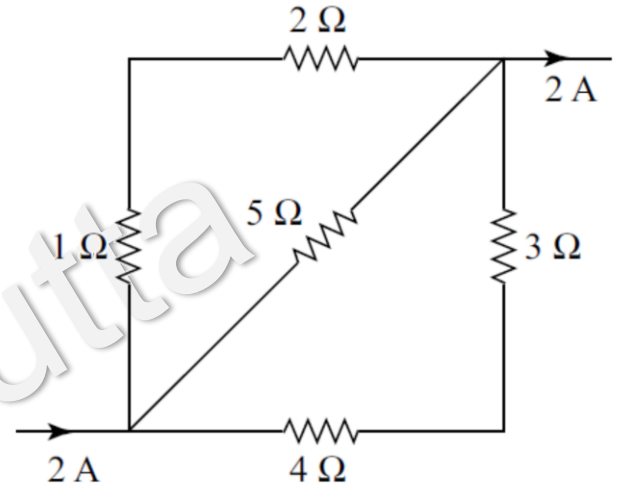
$$\text{or, } -I_2 R_2 - (I_2 - I_1) R_3 - E_2 = 0$$

- If the values of R_1, R_2, R_3, E_1, and E_2, are known, we can calculate the branch currents

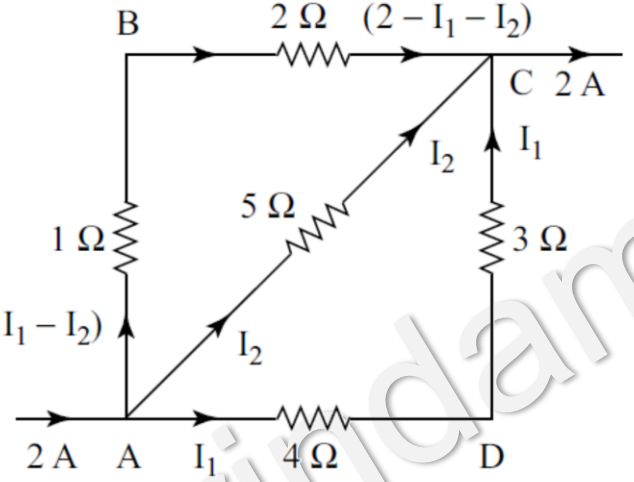
Dr. Arindam Dutta

Problem

Q: Use KCL and KVL to calculate the branch currents in the circuit



A:



KCL applied

Applying KVL at ABCA

$$-1(2 - I_1 - I_2) - 2(2 - I_1 - I_2) + 5I_2 = 0$$

$$3I_1 + 8I_2 = 6$$

Applying KVL at ACDA

$$-5I_2 + 3I_1 + 4I_1 = 0$$

$$7I_1 - 5I_2 = 0$$

$$21I_1 + 56I_2 = 42$$

$$21I_1 - 15I_2 = 0$$

$$I_2 = \frac{42}{71}\text{ A}$$

$$I_1 = \frac{30}{71}\text{ A}$$

$$I_3 = 2 - (I_1 + I_2) = 2 - \left(\frac{30}{71} + \frac{42}{71}\right)$$

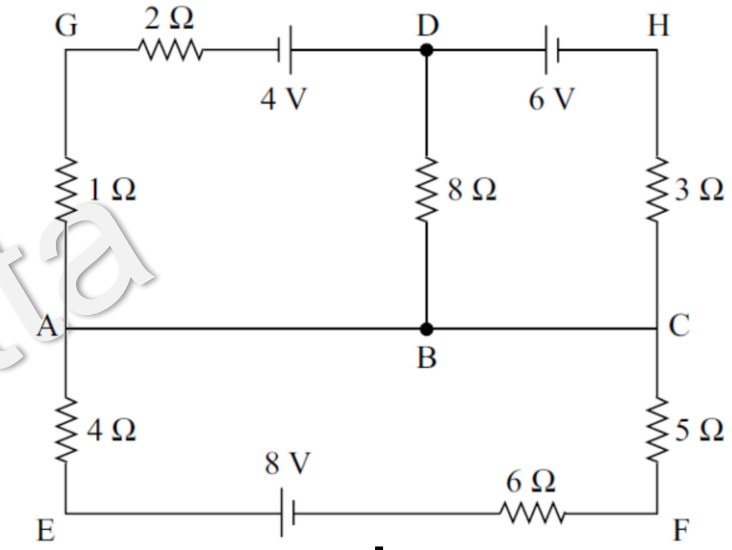
$$= \frac{70}{71}\text{ A}$$

$$I_1 + I_2 + I_3 = 2\text{ A}$$

Dr. Arindam Dutta

Problem

Q: Calculate using KCL & KVL the current flowing through the 8 Ω resistor



A: In EAFE, current I_1 will flow. No current from this loop will flow to the other two loops. Current flowing from E to A is to be the same as the current flowing from A to F.

Applying KVL in GDAG: $-2I_2 + 4 - 8(I_2 + I_3) - I_2 = 0$

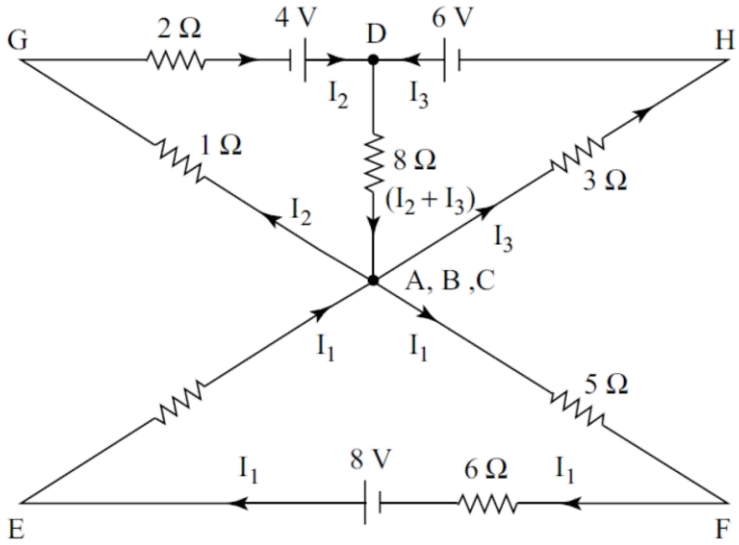
$$11I_2 + 8I_3 = 4$$

Applying KVL in HDAH: $6 - 8(I_2 + I_3) - 3I_3 = 0$

$$8I_2 + 11I_3 = 6$$

$$I_3 = 0.6\ \text{A} \text{ and } I_2 = -0.07\ \text{A}$$

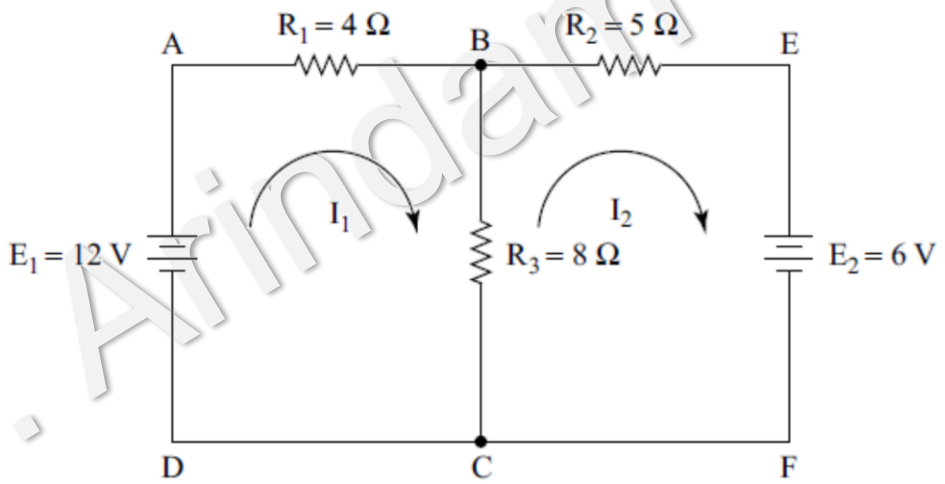
Current through the 8 Ω resistor = $I_2 + I_3 = 0.53$ A



Dr. Arindam Dutta

Maxwell's Mesh Current Method

- KVL is applied to each mesh in terms of mesh currents instead of branch currents
- Mesh currents are assumed to be flowing in the clockwise direction without branching out at junctions
- Applying KVL, the mesh currents are calculated, and subsequently the branch currents can be determined



- Current flowing through R_3 is the algebraic sum of I_1 and I_2

Dr. Arindam Dutta

Maxwell's Mesh Current Method

Applying KVL to ABCDA:

$$12 - 4\,I_1 - 8(I_1 - I_2) = 0$$
$$3\,I_1 - 2I_2 = 3$$

Applying KVL to BEFCB:

$$-5\,I_2 - 6 - (I_2 - I_1)\,8 = 0$$
$$8\,I_1 - 13\,I_2 = 6$$

Solving the two equations: $I_1 = 1.17$ A, $I_2 = 0.26$ A

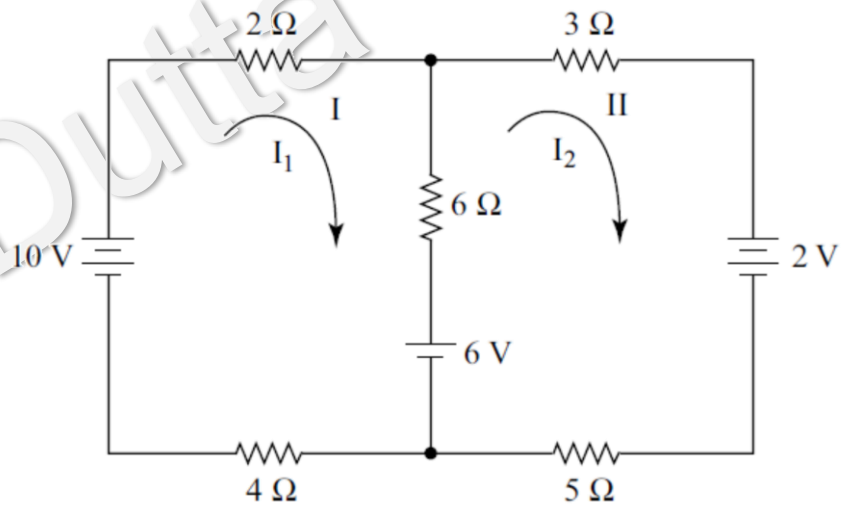
Current flowing through R_3 is $(I_1 - I_2) = 0.91$ A

I_1 is flowing through R_1, I_2 is flowing through R_2 and $(I_1 - I_2)$ is flowing through R_3

Dr. Arindam Dutta

Maxwell's Mesh Current Method

Q: Using the mesh current method calculate the current flowing through the resistors



Applying KVL in Mesh I: $10 - 2\,I_1 - 6\,(I_1 - I_2) - 6 - 4\,I_1 = 0$

$$6I_1 - 3I_2 = 2$$

Applying KVL in Mesh II: $-3I_2 - 2 - 5I_2 + 6 + 6\,(I_1 - I_2) = 0$

$$-6I_1 + 14\,I_2 = 4$$

Solving the two equations: $11\,I_2 = 6$

$$I_2 = \frac{6}{11}\,\text{A}$$

$$I_1 = \frac{2 + 3I_2}{6} = \frac{2 + 3 \times 6/11}{6} = \frac{20}{33}\,\text{A}$$

Current through the 6 Ω resistor is $(I_1 - I_2)$ which is equal to 2 / 33 A.

Dr. Arindam Dutta

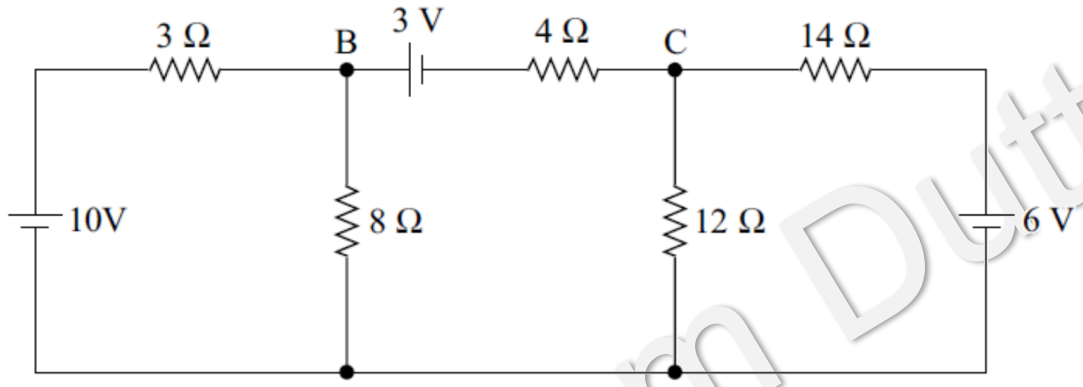
Nodal Voltage Method (Nodal Analysis)

- A reference node in the network is chosen
- The unknown voltages at the other nodes are determined with respect to the reference node
- Currents in all branches are calculated
- Suitable where a network has a number of loops, and hence a large number of simultaneous equations are to be solved

Dr. Arindam Dutta

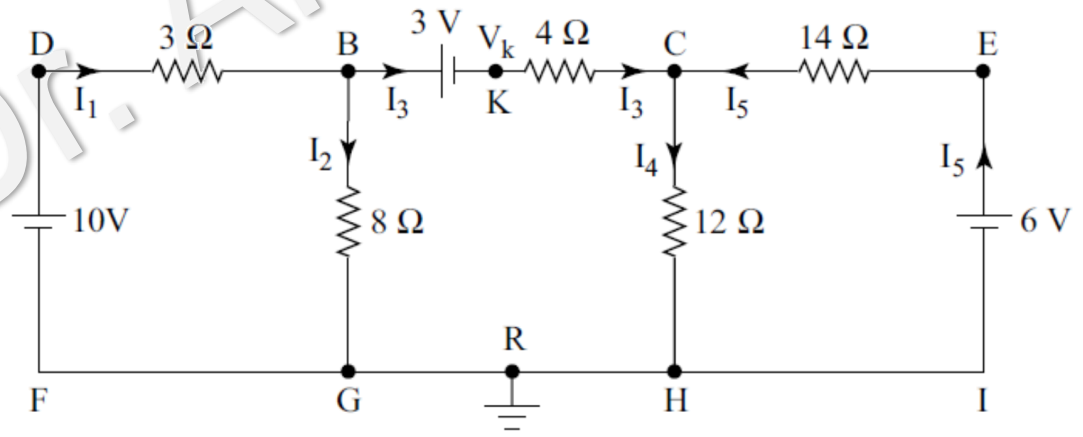
Nodal Voltage Method (Nodal Analysis)

Q: Determine the voltages at nodes B and C and calculate the current through the 8 Ω resistor



A: We will take one reference node (R) at zero potential (ground node or zero potential node).

Generally the node at which maximum branches are meeting is taken as the reference node.



Dr. Arindam Dutta

Nodal Voltage Method (Nodal Analysis)

- Points F, G, R, H, I are at zero reference potential
- Let V_D, V_B, V_C, V_E are the potentials at points D, B, C, and E, respectively
- Applying Ohm's law currents I_1, I_2, I_3, I_4, and I_5 are expressed as

$$\frac{V_D - V_B}{3} = I_1; \quad \frac{V_B}{8} = I_2$$

$$\frac{10 - V_B}{3} = I_1$$

- To find I_3, we assume potential at point k as V_K. So, $V_K + 3 = V_B$

$$I_3 = \frac{V_K - V_C}{4} = \frac{V_B - 3 - V_C}{4}$$

Dr. Arindam Dutta

Nodal Voltage Method (Nodal Analysis)

- Applying KCL at node B:

$$I_1 = I_2 + I_3$$

$$\frac{10 - V_B}{3} = \frac{V_B}{8} + \frac{V_B - V_C - 3}{4}$$

$$17\,V_B - 6\,V_C = 98$$

- Applying KCL at node C:

$$I_4 = I_3 + I_5$$

$$\frac{V_C}{12} = \frac{6 - V_C}{14} + \frac{V_B - V_C - 3}{4}$$

$$21\,V_B - 34\,V_C = 27$$

- Solving the two equations: $V_B = 7.01\ \text{V};\ V_C = 3.537\ \text{V}$
- Current through the 8 Ω resistor: $I_2 = \frac{V_B}{8} = \frac{7.01}{8} = 0.88\ \text{A}$

Dr. Arindam Dutta

EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 1 – Lecture 3

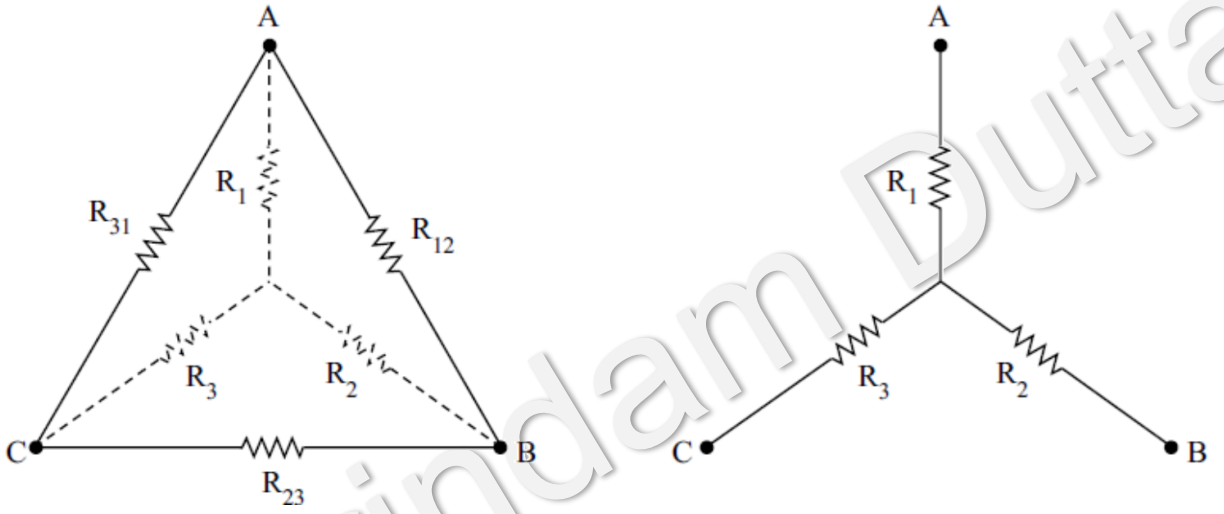
Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **C-509**

Delta – Star Transformation

The two arrangements are equivalent if the resistance measured between any two terminals is the same in both the arrangements



Dr. Arindam Dutta

Delta to Star

$$R_1 = \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$R_2 = \frac{R_{12} R_{23}}{R_{12} + R_{23} + R_{31}}$$

$$R_3 = \frac{R_{23} R_{31}}{R_{12} + R_{23} + R_{31}}$$

Star to Delta

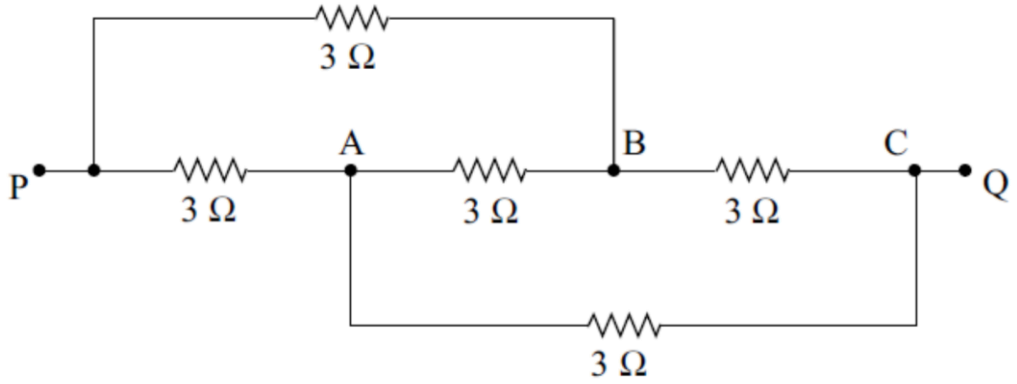
$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

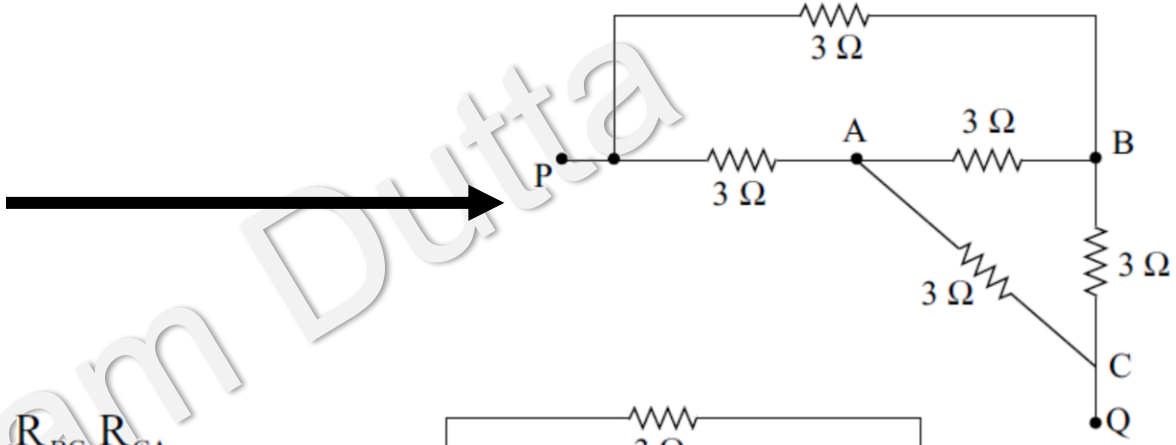
$$R_{23} = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

$$R_{31} = R_3 + R_1 + \frac{R_3 R_1}{R_2}$$

Problem

Q: Calculate the equivalent resistance across P and Q using star-delta conversion.

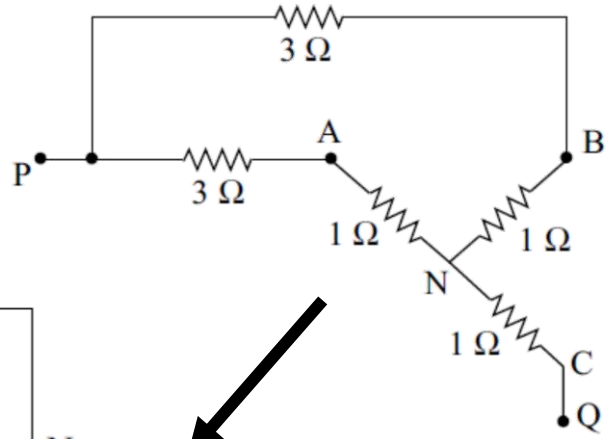


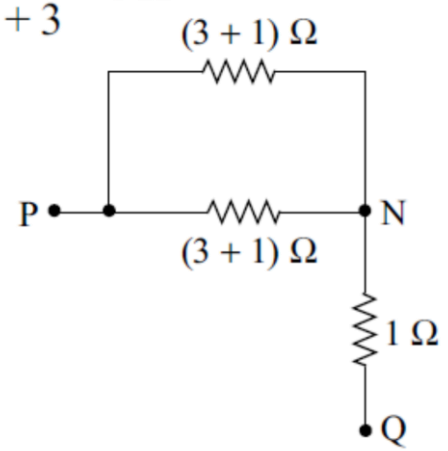


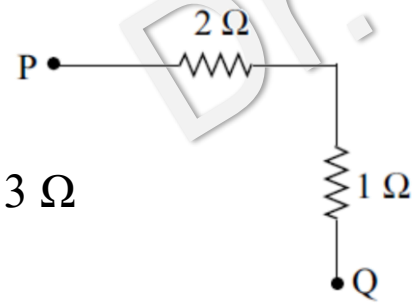
$$R_A = \frac{R_{AB}R_{AC}}{R_{AB}+R_{BC}+R_{CA}} = \frac{3\times 3}{3+3+3} = 1\,\Omega$$

$$R_B = \frac{R_{AB}R_{BC}}{R_{AB}+R_{BC}+R_{CA}} = \frac{3\times 3}{3+3+3} = 1\,\Omega$$

$$R_C = \frac{R_{BC}R_{CA}}{R_{AB}+R_{BC}+R_{CA}} = \frac{3\times 3}{3+3+3} = 1\,\Omega$$





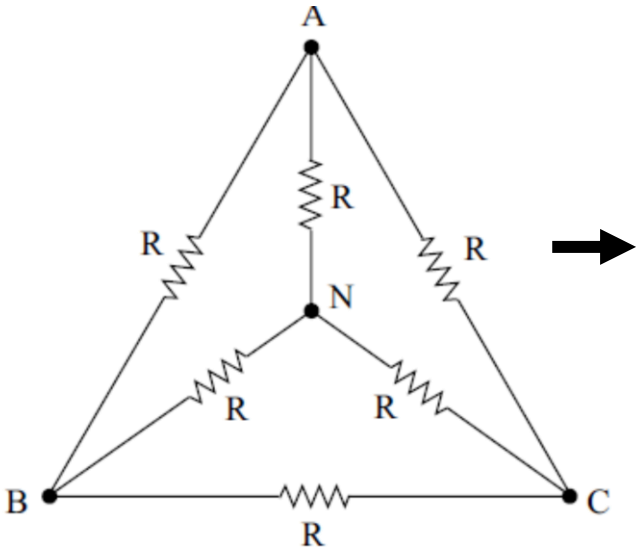


Therefore, $R_{PQ} = 3\ \Omega$

Dr. Arindam Dutta

Problem

Q: Calculate the equivalent resistance across B and C.



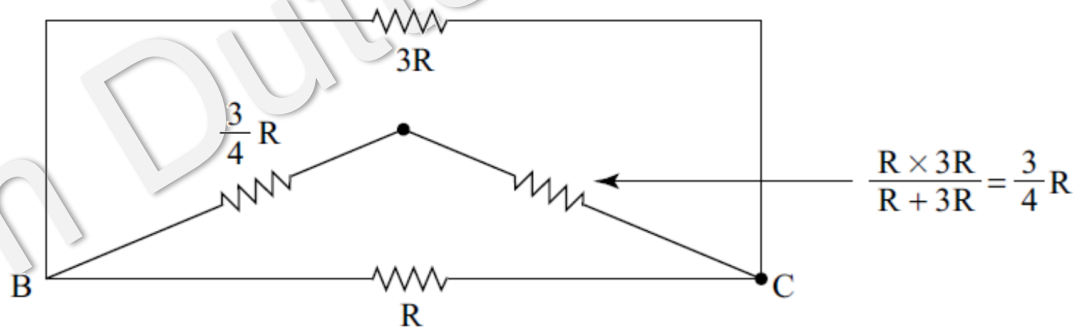
$$R_{AB} = R + R + \frac{R \times R}{R}$$

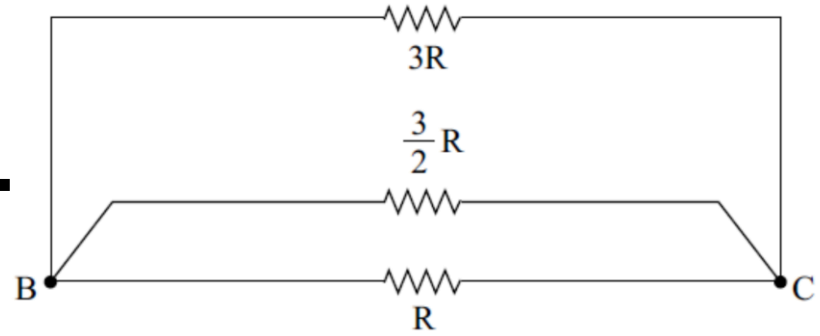
$$= 3R$$

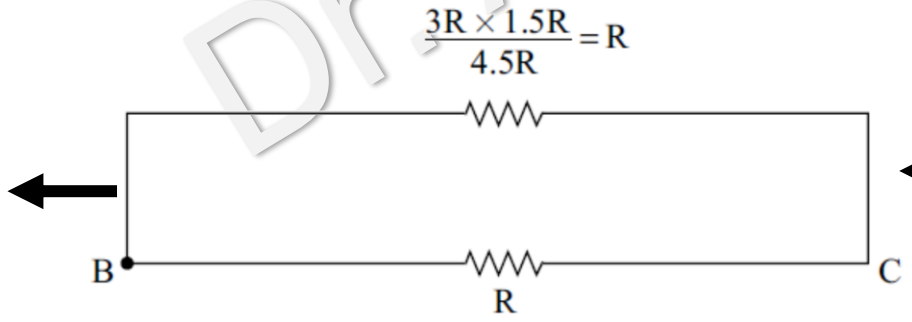
$$R_{CA} = 3R$$

$$R_{BC} = 3R$$

Keeping B and C intact







$R_{BC}(eq) = R/2$

Dr. Arindam Dutta

Network Theorems

- The procedures discussed so far involves solving of number of equations depending upon the complexity of the network
- Many networks require restricted analysis, e.g., finding current through a particular resistor or finding the load resistance at which maximum power will be transferred from the source to the load
- A theorem or method is chosen based on which method will require less time in calculations for the circuit
- Theorems:
  - Superposition theorem
  - Thevenin's theorem
  - Norton's theorem
  - Maximum power transfer theorem

Dr. Arindam Dutta

Superposition Theorem

- An electrical circuit may contain more than one source of supply (voltage or current source)
- The effect of each source is calculated separately and the combined effect of all the sources are taken into consideration
- The superposition theorem states that

  In a linear network containing more than one source, the current flowing in any branch is the algebraic sum of currents that would have been produced by each source taken separately, with all the other sources replaced by their respective internal resistances.*

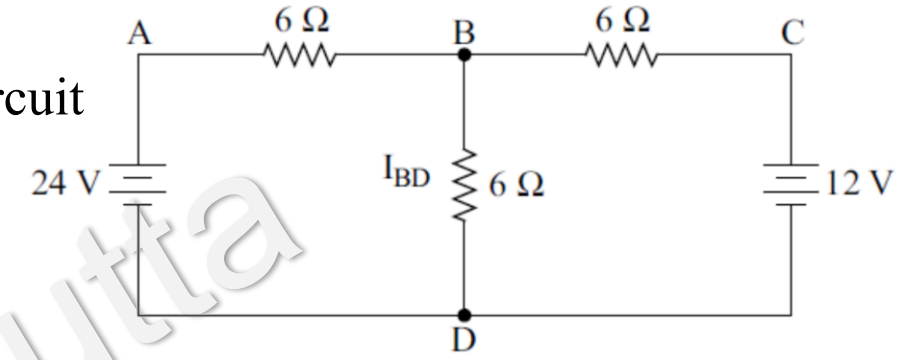
  In case the internal resistance of a source is not provided, the voltage sources will be short circuited and current sources will be open circuited.

*A circuit whose parameters are constant, i.e., they do not change with application of voltage or current is called a linear circuit.

Dr. Arindam Dutta

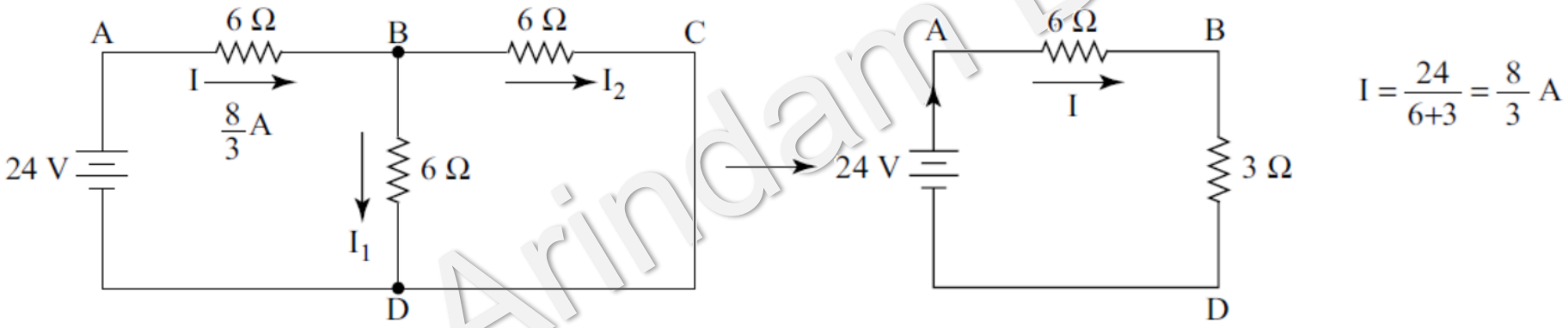
Superposition Theorem

Q: Using the superposition theorem find the value of current, I_{BD} in the circuit



A: Each source is considered separately to calculate the current flowing through the branch BD

First the 24 V source is taken by short circuiting the 12 V source



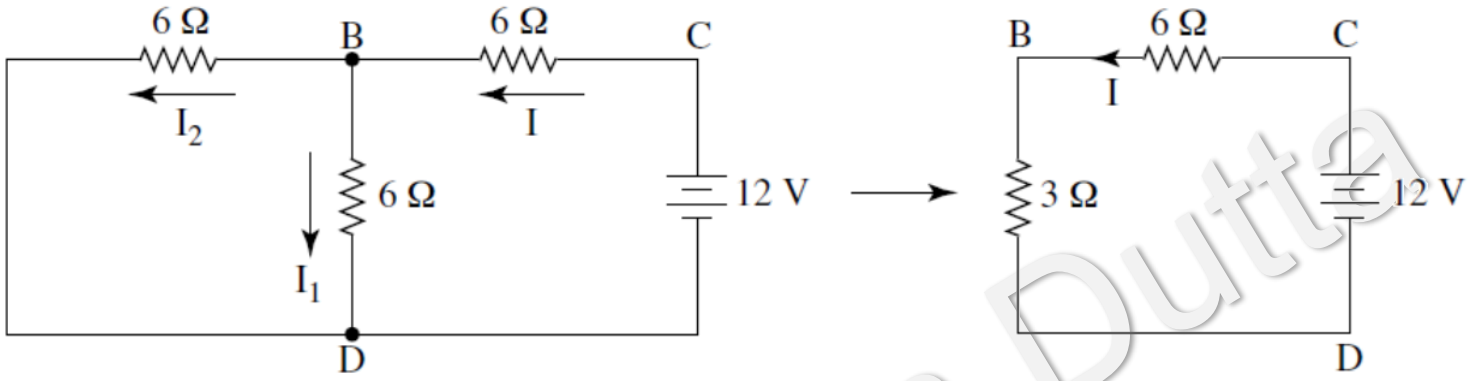
$$I = \frac{24}{6+3} = \frac{8}{3}\ A$$

The current flowing from the battery is calculated as I = 8/3 A. This current gets divided into two parts as I_1 and I_2. Current through the resistor across BD is I_1. To find I_1 we can use the current division rule as:

$$I_1 = I\frac{R_2}{R_1 + R_2} = \frac{8}{3} \times \frac{6}{6+6} = \frac{4}{3}\ A$$

Dr. Arindam Dutta

Superposition Theorem



Now, consider the 12 V source and short circuit the 24 V source. The current supplied by the 12 V source is calculated as I = 12 / (6 + 3) = 4/3 A.

This current gets divided into I_1 and I_2. Current I_1 is calculated using the current division rule as

$$I_1 = I \times \frac{R_2}{R_1 + R_2} = \frac{4}{3} \times \frac{6}{6+6} = \frac{2}{3} \text{A}$$

To determine the current flowing through the resistor across BD, the combined effect of the two voltage sources will be taken.

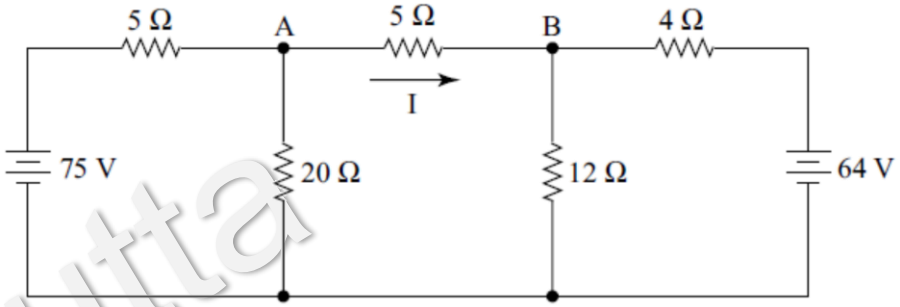
Therefore, I_{BD} = I_1 due to 24 V source + I_1 due to the 12 V source = 2 A, flowing from B to D

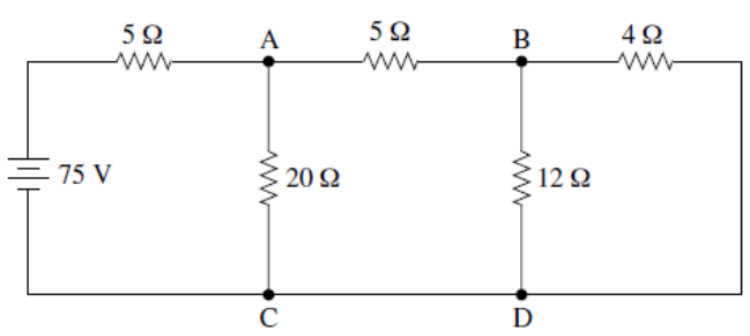
Dr. Arindam Dutta

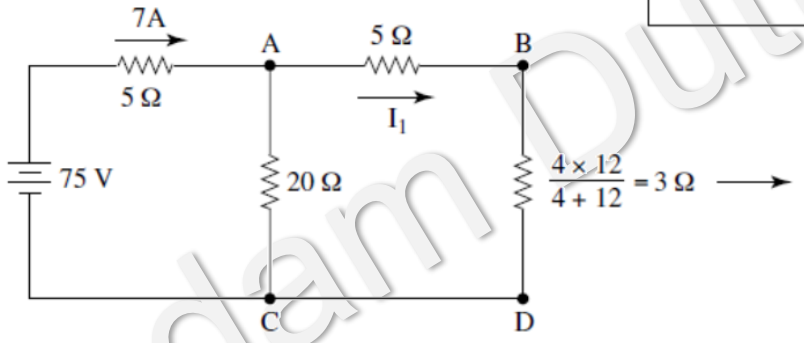
Superposition Theorem

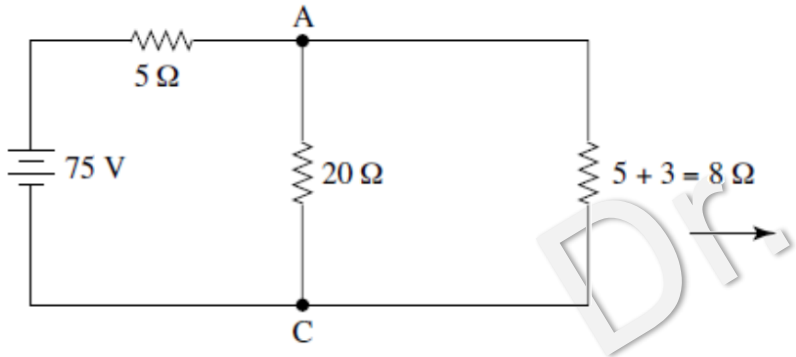
Q: Calculate the current, I using the superposition theorem.

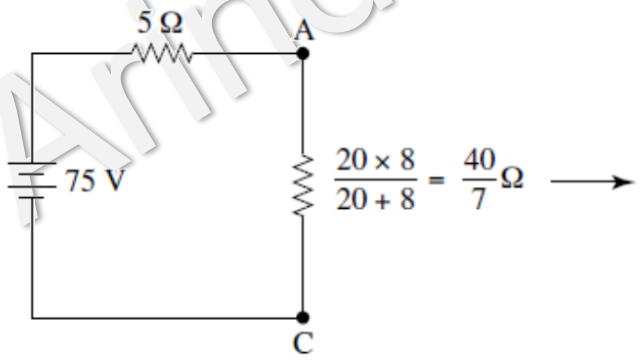
A: We will consider the 75 V source first and short circuit the 64 V.

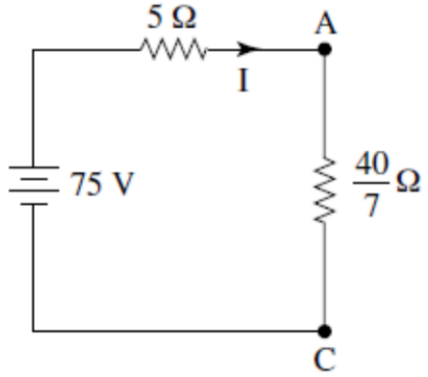












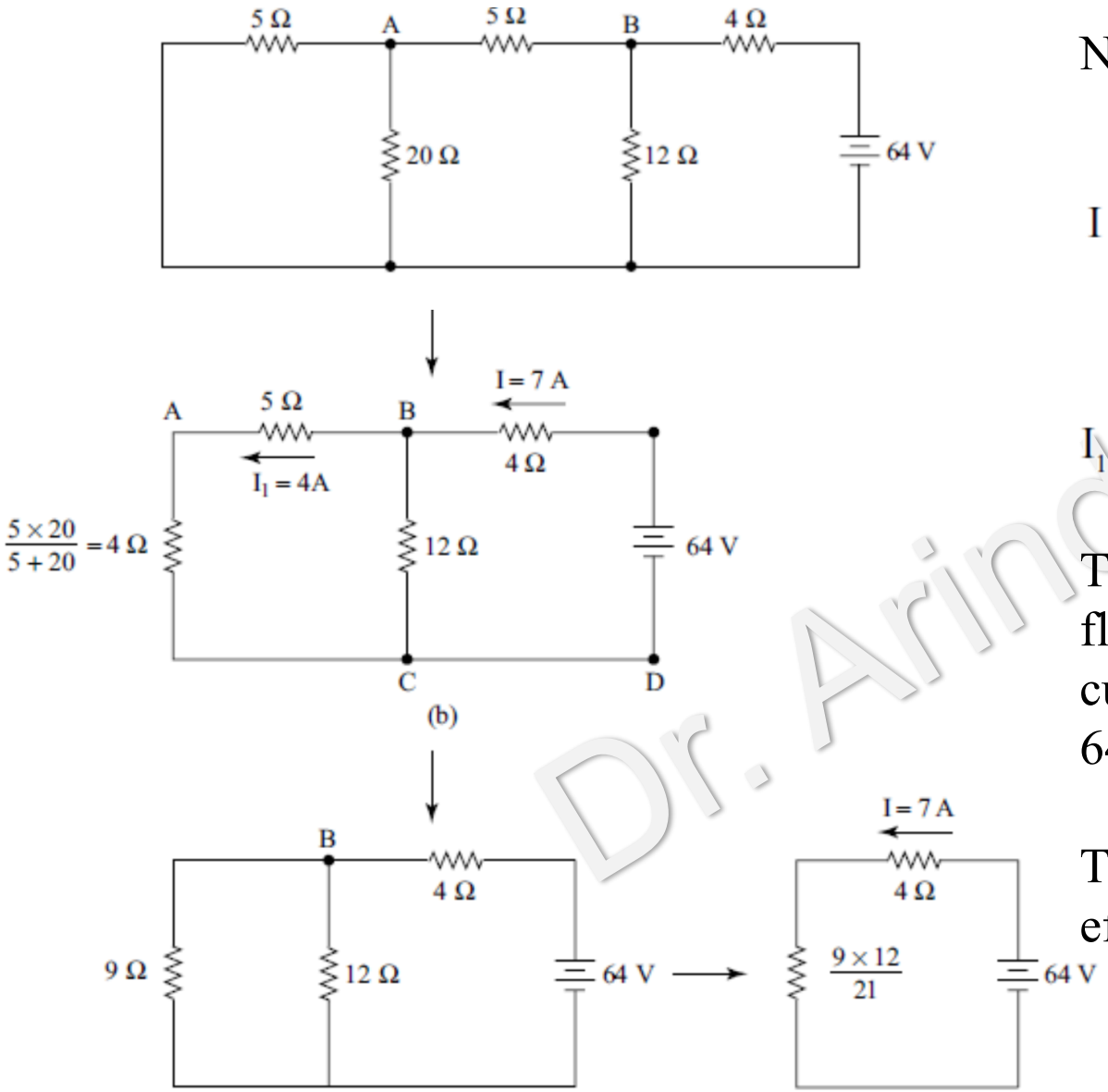
$$I = \frac{75}{5 + \frac{40}{7}} = \frac{75}{\frac{75}{7}} = 7\ A$$

The battery current calculated is 7 A and current through the 5 Ω resistor across A and B is:

$$I_1 = 7 \times \frac{20}{20 + 5 + 3} = 5A$$

Dr. Arindam Dutta

Superposition Theorem



Now, short circuiting the 75 V source:

$$I = \frac{64}{4 + 108/21} = \frac{64}{84 + 108/21} = \frac{64 \times 21}{192} = 7\text{A}$$

$$I_1 = I \times \frac{12}{5 + 4 + 12} = 7 \times \frac{12}{9 + 12} = 4\ \text{A}$$

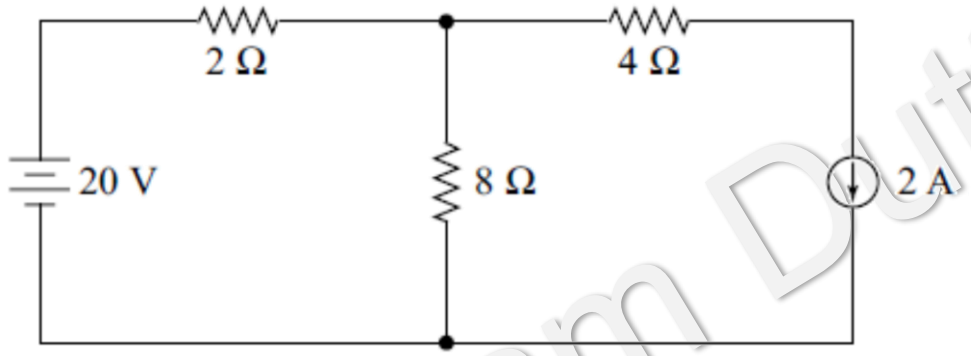
Through the resistor across terminals AB, current of 5 A flows from A to B due to voltage source of 75 V and a current of 4 A flows from B to A due to voltage source of 64 V.

The net current I_{AB} is equal to $I_{AB} = 5 - 4 = 1$ A when the effect of both the voltage sources are superimposed.

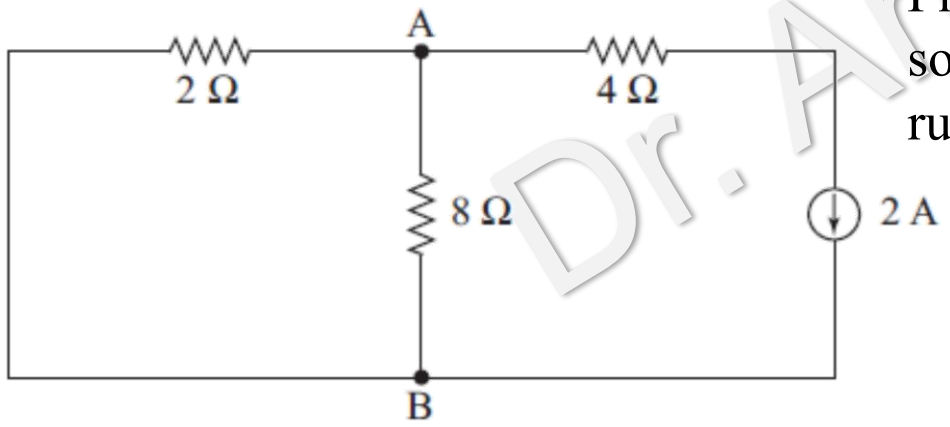
Dr. Arindam Dutta

Superposition Theorem

Q: Determine current through the 8 Ω resistor in the network using the superposition theorem.



A:



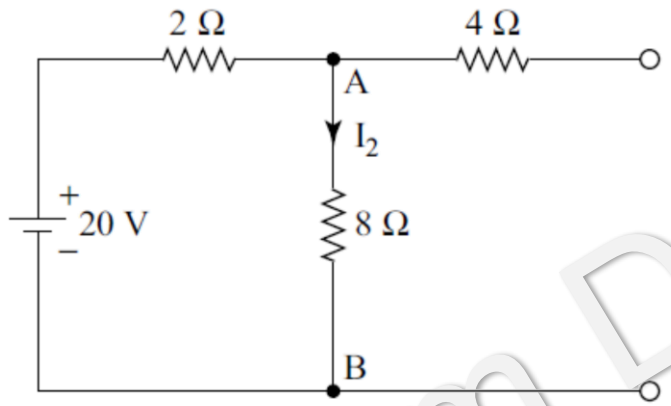
First the effect of current source will be considered. The voltage source is replaced by a short circuit. Using the current division rule, we determine current I_1 flowing from B to A as

$$I_1 = 2 \times \frac{2}{2+8}$$

$$= 0.4\ \text{A}$$

Dr. Arindam Dutta

Superposition Theorem



Now if only the voltage source is considered, the current source has to be open circuited. Current flowing through the 8 Ω resistor is determined as

$$I_2 = \frac{20}{2+8} = 2\ \text{A}$$

Hence total current in the 8 Ω resistor from A to B is resistor: $I_2 - I_1 = 1.6$ A

Dr. Arindam Dutta

EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 1 – Lecture 4

Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **C-509**

Thevenin's Theorem

- The whole circuit across the terminals of the resistor, through which current flow is to be calculated, is converted into a voltage source with an internal resistance
- The voltage is the open circuit voltage of the network across the terminals
- Internal resistance is the equivalent resistance of the whole circuit across the open-circuited terminals by replacing the voltage sources by their internal resistances
- The current through the resistor R:

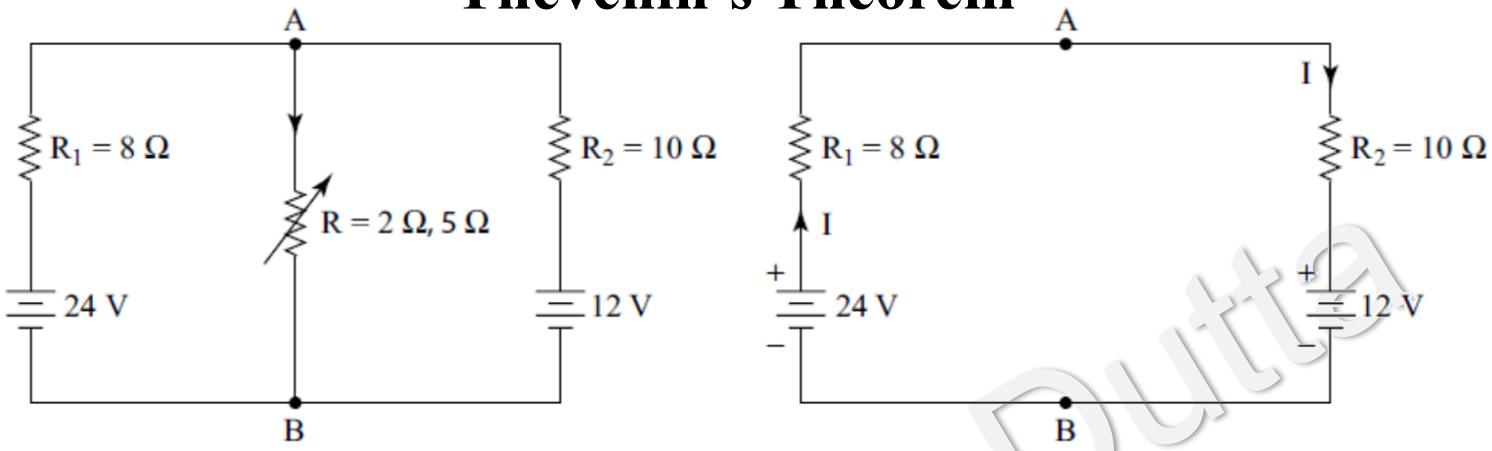
$$\mathbf{I = V_{oc} / (R_{eq} + R)}$$

V_{OC} is the open-circuit voltage across the terminals of resistor R (when R is removed from the circuit);

R_{eq} is the equivalent circuit resistance across the terminals of R

Dr. Arindam Dutta

Thevenin's Theorem



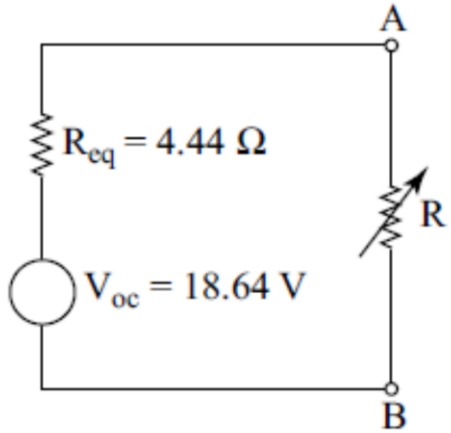
To calculate the current through the variable resistor R, the first step would be to take away the resistor R and calculate the V_{OC} across terminals A and B.

Applying KVL in the circuit: $+24 - 8\,I - 10\,I - 12 = 0$

$$I = 0.67\ A$$

Approaching from B to A: $V_{OC} = 24 - 8 \times 0.67 = 18.64\ V$

Thevenin's equivalent circuit



Now, we calculate R_{eq} across A and B by short circuiting the voltage sources or by replacing them by their internal resistances.

Across terminals AB we find two resistances of 8 Ω and 10 Ω connected in parallel.

R_{eq} across terminals A and B = 4.44 Ω

Dr. Arindam Dutta

Thevenin's Theorem

Current through R with its variable values i.e., for R = 2 Ω and R = 5 Ω are calculated as

$$I = \frac{V_{OC}}{R_{eq} + R} = \frac{18.64}{4.44 + 2} = 2.89\,A$$

$$I = \frac{V_{OC}}{R_{eq} + R} = \frac{18.64}{4.44 + 5} = 1.97\,A$$

Thus, Thevenin's theorem states that:

Any two terminals of an electrical network consisting of active and passive elements (i.e., voltage sources and resistors) can be replaced by an equivalent voltage source and an equivalent series resistance.

The voltage source is the open-circuit voltage between the terminals caused by the active network. The series resistance is the equivalent resistance of the whole circuit across the terminals looking to the circuit from the two terminals with all the sources of EMF short circuited.

Dr. Arindam Dutta

Thevenin's Theorem

The **steps** involved in applying Thevenin's theorem in calculating current in a circuit component are as follows:

1. Remove the resistance from the circuit terminals through which current is to be determined.

2. Determine the open-circuit voltage that would be appearing across the circuit terminals wherefrom the resistance has been removed. This is called V_{OC}.

3. Calculate the equivalent resistance of the whole circuit across the terminals after replacing the sources of EMFs by their internal resistances (or by simply short–circuiting them if internal resistance is not provided or not known) and by keeping the current sources open–circuited (considering having infinite resistance).

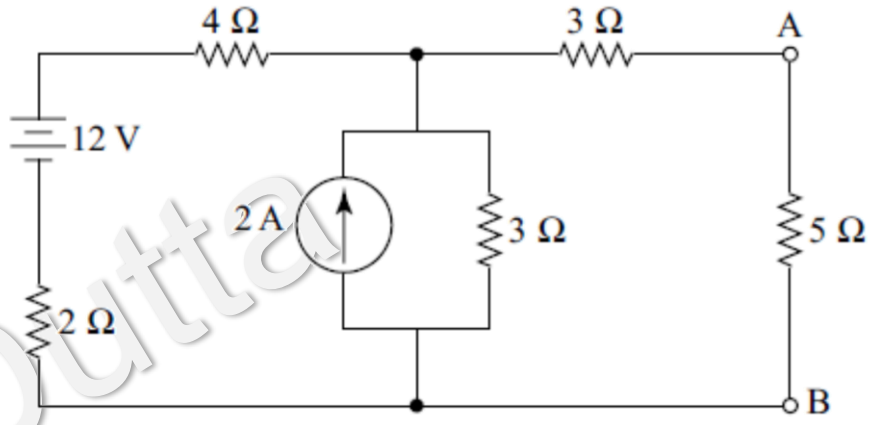
4. Draw the Thevenin's equivalent circuit with V_{OC} as the voltage source, R_{eq} as the internal resistance of the voltage source, and R is the load resistance connected across the voltage source.

5. Calculate the current through R using the relation $\mathbf{I = V_{oc} / (R_{eq} + R)}$

Dr. Arindam Dutta

Thevenin's Theorem

Q: Use Thevenin's theorem to calculate the current through the 5 Ω resistor.

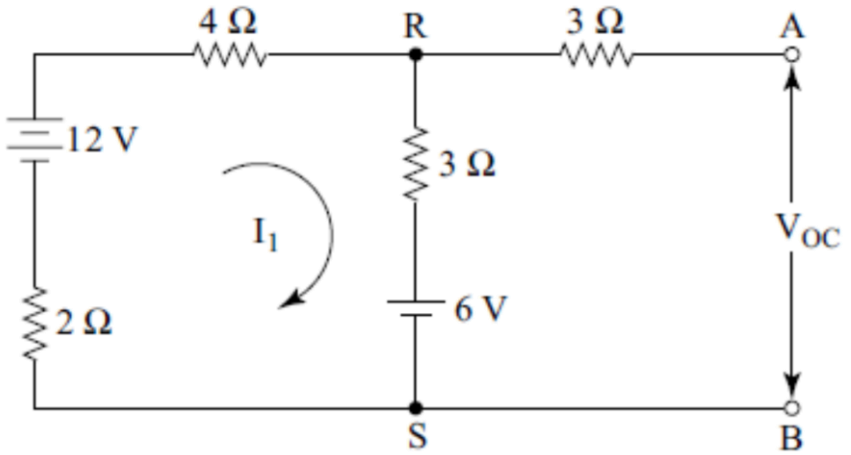


A: We convert the current source into its equivalent voltage source and take away the 5 Ω resistor from A and B.

Using KVL in the loop:

$-2I_1 + 12 - 4I_1 - 3I_1 - 6 = 0 \Rightarrow I_1 = 2/3$ A

Voltage across point S and R is calculated by moving upwards from point S to point R.



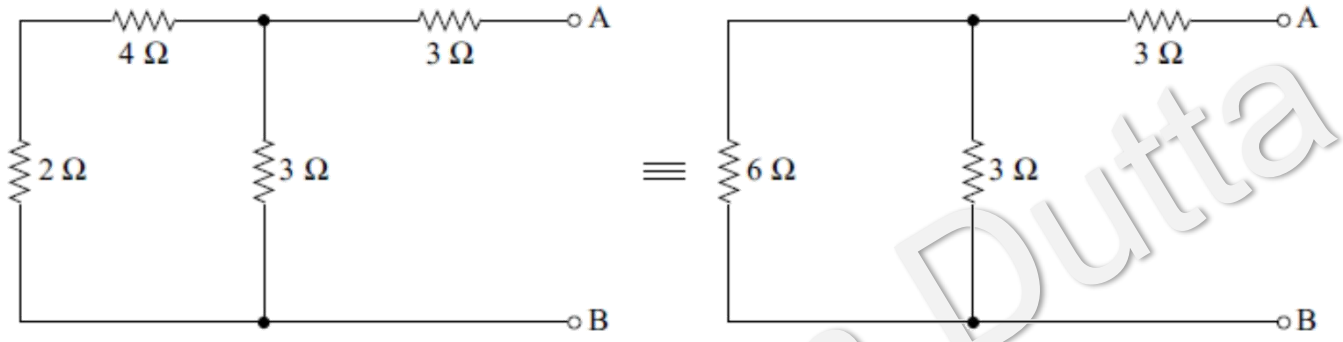
Considering the voltage rise we get, $+6 \text{ V} + 3 \times (2/3) \text{ V} = +8 \text{ V}$

As there is no current flowing in the 3 Ω resistor between R and A, potential across A w.r.t. B is 8 V, A being at higher potential than B.

Dr. Arindam Dutta

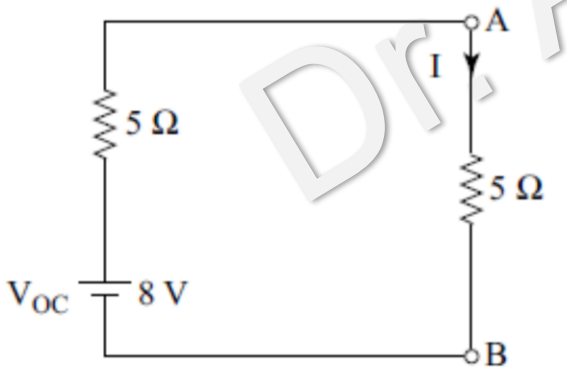
Thevenin's Theorem

R_{eq} is calculated as:



$$R_{eq} = R_{AB} = 3 + (3 \times 6 / 9) = 5\ \Omega$$

Thevenin's equivalent circuit:

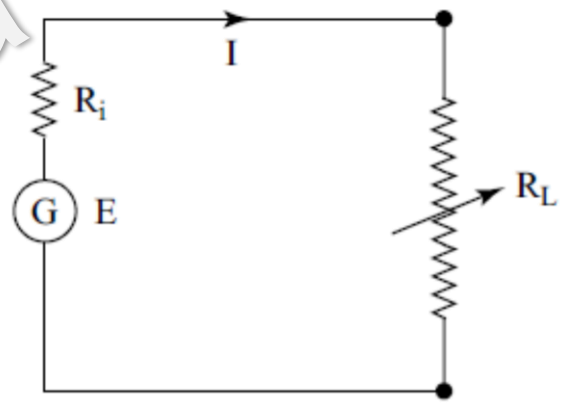


$$I = 8 / (5 + 5)\ A = 0.8\ A$$

Dr. Arindam Dutta

Maximum Power Transfer Theorem

- Power is supplied from a source to a load. Let us consider a generator with internal resistance, R_i supplying power to a load resistance, R_L. Maximum power transfer theorem tells us at what load (i.e., at what value of R_L) maximum power will be transferred from the source to the load.



- $I = E / (R_i + R_L)$

- Let power consumed by or delivered to the load be P_L. Since power = I^2R

$$P_L = I^2 R_L = \left(\frac{E}{R_i + R_L}\right)^2 R_L$$

- Since R_L is variable, for determining the condition for maximum power transfer from the source to the load, we will differentiate P_L with respect to R_L.

$$\frac{dP_L}{dR_L} = 0$$

$$\frac{dP_L}{dR_L} = \frac{d}{dR_L}\left[\frac{E^2 R_L}{(R_L + R_i)^2}\right] = 0 \qquad \text{or,} \qquad \frac{d}{dR_L} E^2 \left[\frac{R_L}{(R_L + R_i)^2}\right] = 0$$

Dr. Arindam Dutta

Maximum Power Transfer Theorem

$$\frac{d}{dR_L} E^2 [R_L (R_L + R_i)^{-2}] = 0$$

$$E^2 \left[\frac{1}{(R_L + R_i)^2} - \frac{2R_L}{(R_L + R_i)^3} \right] = 0$$

$$2R_L (R_L + R_i)^2 = (R_L + R_i)^3$$

$$2R_L = R_L + R_i$$

$$R_L = R_i$$

Thus, maximum power transfer theorem states that

In a dc network maximum power will be consumed by the load or maximum power will be transferred from the source to the load when the load resistance becomes equal to the internal resistance of the network as viewed from the load terminals.

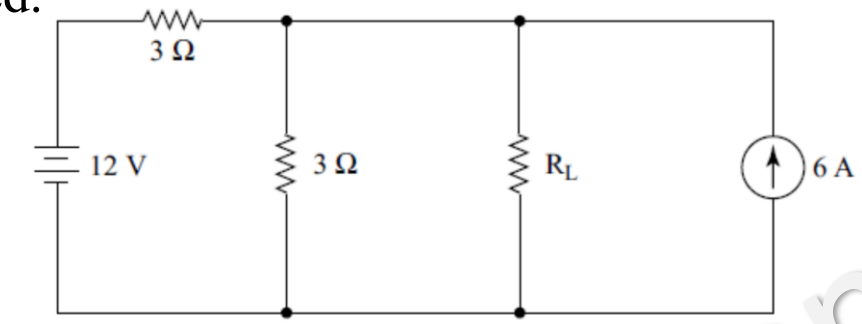
The value of maximum power, when $R_L = R_i$ is $P_L(\max) = \frac{E^2 R_L}{(R_L + R_L)^2}$ (since $R_i = R_L$) $= \frac{E^2}{4R_L} = \frac{E^2}{4R_i}$

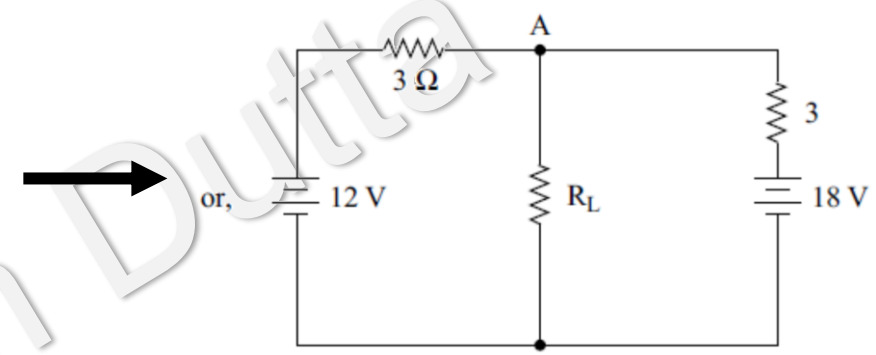
When a complex dc network is to be analyzed for maximum power transfer, the circuit can first be converted into a voltage source with one internal resistance by applying **Thevenin's theorem**.

Dr. Arindam Dutta

Maximum Power Transfer Theorem

Q: Calculate the value of load resistance, R_L for which maximum power will flow to the load. Also calculate the maximum power transfer efficiency, i.e., the power transmission efficiency when maximum power is transferred.

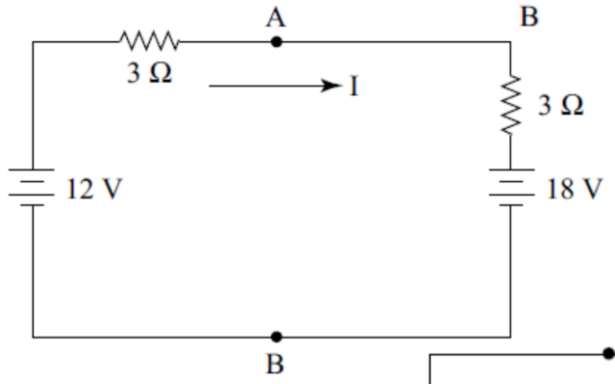




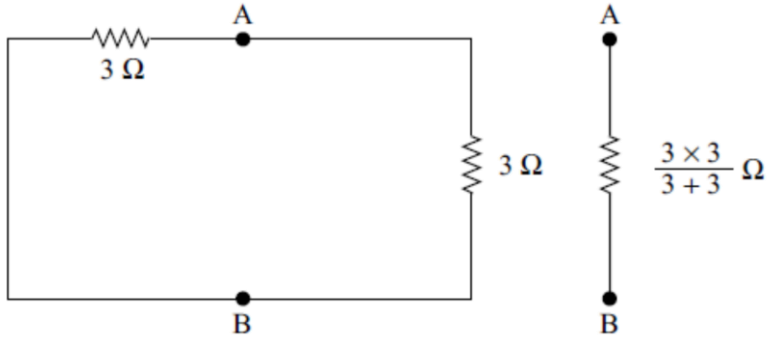
To calculate the open circuit voltage across A and B, we apply KVL :

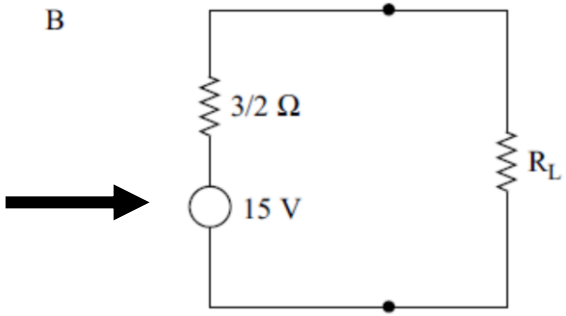
$+12 - 3I - 3I - 18 = 0;\ I = -1$ A

$V_{OC} = 18 + 3 \times (-1) = 15$ V



Short-circuiting voltage sources,
R_{eq} (AB) = 3/2 Ω





Thevenin's Equivalent Circuit

Dr. Arindam Dutta

Maximum Power Transfer Theorem

For maximum power transfer, $R_L = 3/2\ \Omega$

$P_L\ (max) = E^2 / 4\ R_L = 15^2 / (4 \times 3/2)\ W = 37.5\ W$

Power transfer efficiency, $$\eta = \frac{\text{Power supplied to the load}}{\text{Total power supplied by the source}}$$

$$= \frac{I^2 R_L}{I^2 R_i + I^2 R_L} = \frac{R_L}{R_i + R_L}$$

$$= \frac{1}{1 + (R_i / R_L)}$$

when $R_i = R_L$,

$$\%\eta = \frac{1}{1+1} \times 100 = 50 \text{ per cent}$$

This shows that for maximum power transfer conditions, the power transfer efficiency is 50% only.

The maximum power transferred from the source to the load and also the power transfer efficiency are important in practical applications. If we change R_L, the value of power transferred and the transmission efficiency will change.

Dr. Arindam Dutta

Norton's Theorem

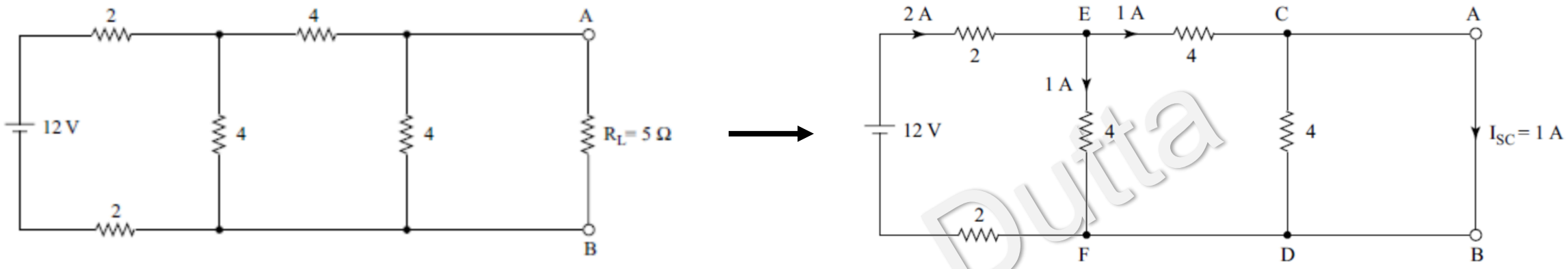
- In contrast to Thevenin's theorem, here the network is converted into a constant current source and a parallel resistance across the terminals of the resistance through which current has to be calculated

- Norton's theorem states that:

  Any two terminal networks consisting of voltage sources and resistances can be converted into a constant current source and a parallel resistance. The magnitude of the constant current is equal to the current which will flow if the two terminals are short circuited and the parallel resistance is the equivalent resistance of the whole network viewed from the open-circuited terminals after all the voltage and current sources are replaced by their internal resistances

Dr. Arindam Dutta

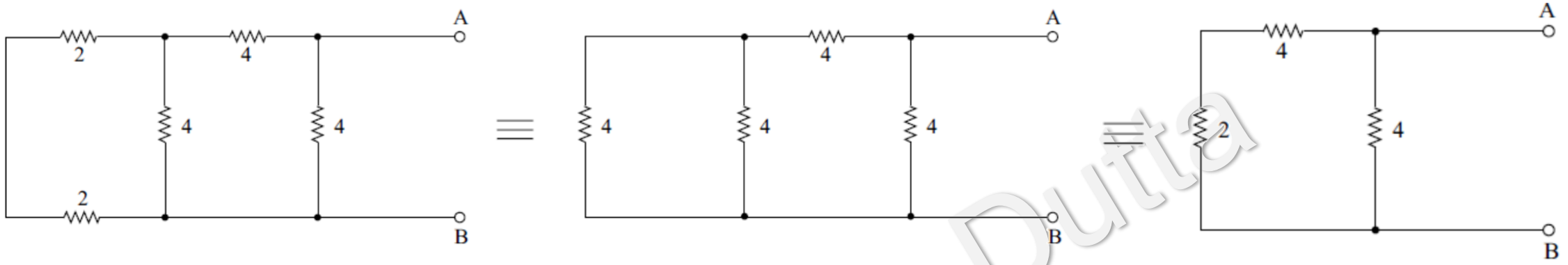
Norton's Theorem



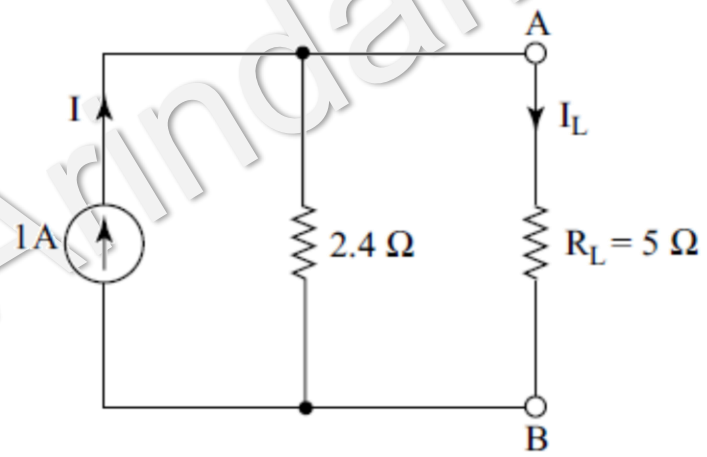
- To calculate the current that would flow through the load resistance, R_L = 5 Ω, the first step is to remove the load resistance and short circuit the terminals AB and calculate I_{SC}
- By observation we see that when terminals A and B are shorted, terminals C and D also get shorted
- Two 4 Ω resistances get connected in parallel across terminals EF, their equivalent resistance become 2 Ω
- Thus the total current, I supplied by the battery = 12 / (2 + 2 + 2) A = 2 A
- At E, 2 A gets divided equally; 1 A going in branch EF and 1 A to branch EC (CD being shorted). At C, 1 A current will flow through the short-circuited path provided between terminals A and B. Therefore, I_{SC} = 1 A

Dr. Arindam Dutta

Norton's Theorem



- The resistance of the network viewed from AB when the battery is short circuited: $R_{eq} = 6 \times 4 / 10 = 2.4\ \Omega$



Norton's equivalent circuit

- Now, using current divider rule, the current through the load resistance: $I_L = 1 \times 2.4 / (2.4 + 5) = 0.32$ A

Dr. Arindam Dutta

Norton's Theorem

The **steps** involved in applying Norton's theorem in calculating current in a circuit component are as follows:

1. Remove the resistance and short circuit the terminals through which current is to be determined.

2. Determine the short-circuit current that would be flowing across the circuit terminals wherefrom the resistance has been removed. This is called I_{SC}.

3. Open circuit the terminals under consideration and calculate the equivalent resistance of the whole circuit across the terminals after replacing the sources of EMFs by their internal resistances (or by simply short–circuiting them if internal resistance is not provided or not known) and by keeping the current sources open–circuited (considering having infinite resistance).

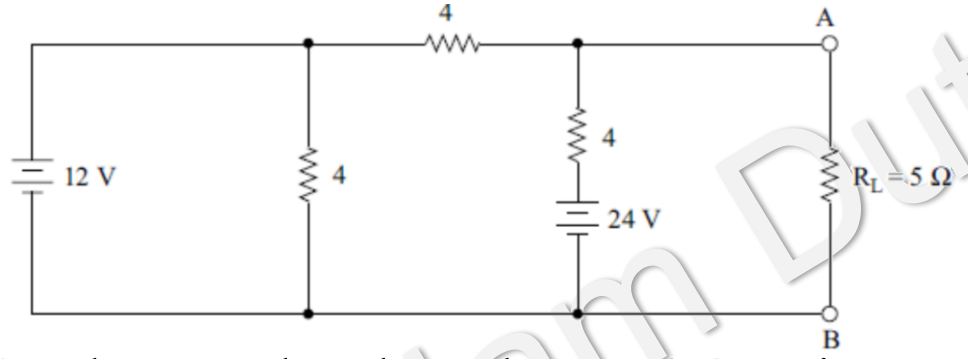
4. Draw the Norton's equivalent circuit with I_{SC} as the current source, R_{eq} as the internal resistance of the current source in parallel, and R as the load resistance connected in parallel to them.

5. Calculate the current through R using the current divider formula $\mathbf{I = I_{SC} \times R_{eq} / (R_{eq} + R)}$

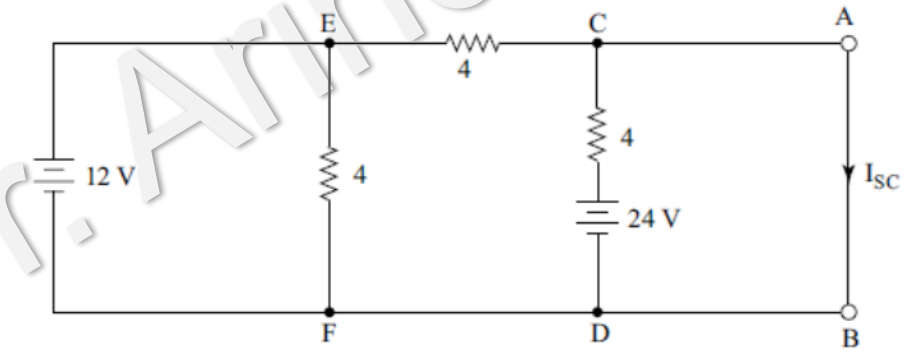
Dr. Arindam Dutta

Norton's Theorem

Q: Using Norton's theorem, calculate the current flowing through the load resistance connected across the terminals A and B.



A: When A and B are shorted, C and D are also shorted. Two 4 Ω resistances are seen connected in parallel across E and F. The current from the battery of 12 V is 12/2 = 6 A.

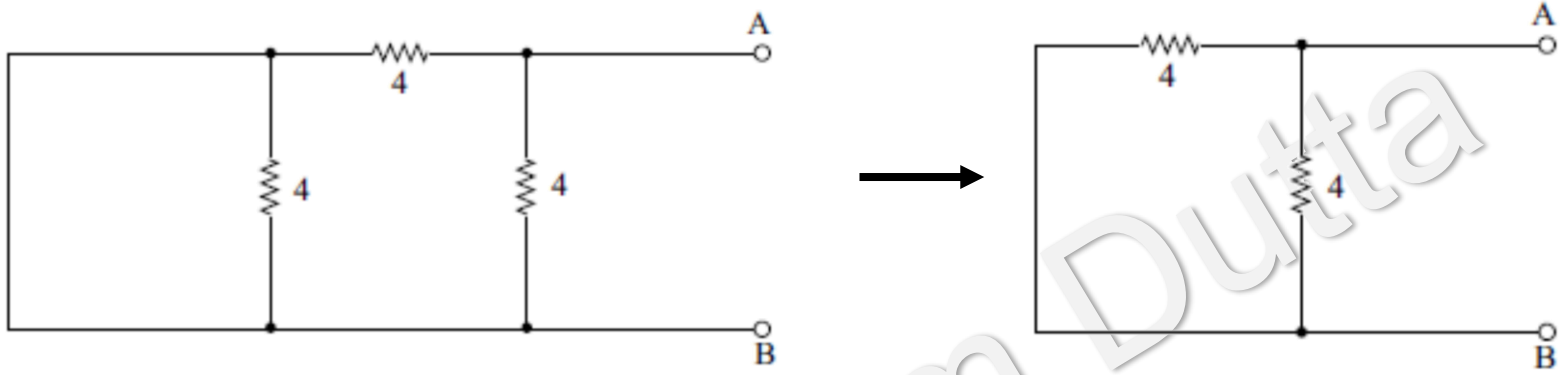


6 A will get divided into 3 A each at branch EF and ECABDF. Thus, I_{SC} due to the 12 V battery source is 3 A. I_{SC} due to the 24 V battery is calculated by considering the loop DCABD which is 24/4 = 6 A. Thus, total I_{SC} due to both the sources of EMF is 3 + 6 = 9 A.

Dr. Arindam Dutta

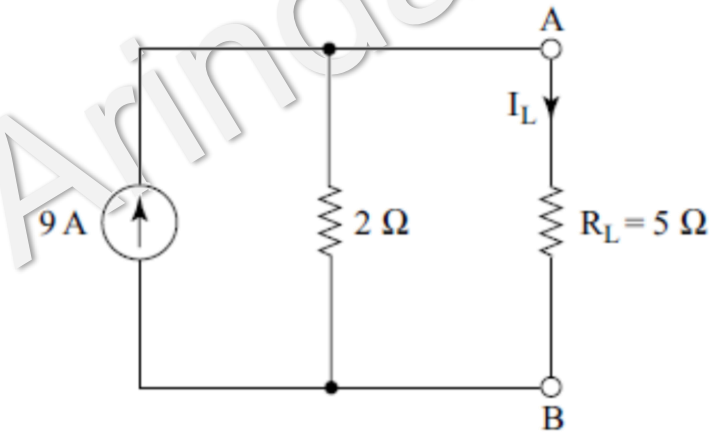
Norton's Theorem

The equivalent resistance of the network across A and B (after short circuiting the sources of EMFs) is:



$R_{eq} = (4 \times 4) / (4 + 4) = 2\ \Omega$

Norton's equivalent circuit:



Therefore, $I_L = 9 \times 2 / (2 + 5) = 18/7$ A

Dr. Arindam Dutta